Month-Shift-Aware Reconstruction, Queue Forecasting and Demand Recovery on TrafficFlowBench: Team *vacucity*

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***Abstract*—TrafficFlowBench asks for four things on ten synthetic Californian freeway corridors: reconstruct masked speed and flow (Task 1), forecast which links are queued in the next 30 minutes (Task 2), stay consistent with traffic physics (Task 3, scored on the Task 1 answer) and recover path flows from link counts (Task 4). The difficulty is not the model class but a shift: the two scored months (March and April 2031) are a different random realisation of the generator (with an organiser-held seed) than the nine training months, and the scoring itself adds mechanics that never occur in training, most notably 90-minute corridor-wide blackouts after every Task 2 forecast origin. Our pipeline is built around that shift. It relies on signals that survive it (flow at the origin relative to bottleneck capacity, the observation row at the origin, corridor structure), adapts to the new scenario only with data the rules allow (self-supervised month adaptation and month-specific recalibration for the offline Task 1; April-only queue models fine-tuned on March for the causal Task 2), and encodes documented evaluation structure (onset at T+30, conditioning on the official window-selection rules, blackout-gap specialists). Changes were validated with proxies that we calibrated against the leaderboard (pseudo and replica windows, hide-20% frames, a blackout ruler, half-month cross-fitting), and the later stages of the model chain were decided by rules registered in advance; every deviation is disclosed. The last step, taken without retraining, adds two guards to the Task 2 queue rows of the model chain's last build, `V84`: a spatio-temporal guard flips 94 cells, and a history guard, which compares each window's history with training-period analogues, turns 10 further cells from not queued to queued. The two selected final files, `V86` (the spatio-temporal guard) and `V87` (both guards), score 0.90359 and 0.90368 on the public leaderboard; on the private leaderboard `V86`, the better of the two, scores 0.89962 (7th of 802 teams on the Kaggle private leaderboard). We document what transferred, what did not, an error budget for all four tasks, every compliance disclosure, and a single-entry reproduction package.**

***Index Terms*—traffic state estimation, queue forecasting, distribution shift, LWR conservation, origin–destination estimation, gradient boosting, U-Net, BiLSTM, reproducibility**

I. Introduction

Freeway operators rarely see the whole corridor: detectors drop out, data are withheld, and decisions have to be taken minutes ahead of a breakdown. TrafficFlowBench [1], [2] turns this into four scored tasks on ten directional corridors of five Californian freeway families at 5-minute resolution. Task 1 (weight 0.35) reconstructs speed and flow at masked detector cells; Task 2 (0.30) forecasts the binary queue state of every link for T+5, ..., T+30 minutes in onset and ongoing windows; Task 3 (0.15) scores the physical consistency of the Task 1 answer against a fundamental diagram and the Lighthill–Whitham–Richards (LWR) conservation law; Task 4 (0.20) recovers path flows from link counts and a weak prior (origin–destination matrix estimation, ODME).

The central difficulty is shift, not capacity. Train covers June 2030 to February 2031; the public leaderboard scores March 2031 and the private one April 2031, "a different month of the same scenario" as March, generated from a random seed held by the organisers. In practice different bottlenecks break first, per-link sensor biases change from month to month, and models that learn when and where congestion usually appears in training lose on the scored months (Section III). On top of that the evaluation adds mechanics that training never shows: every scored Task 2 window blanks the whole corridor for 90 minutes after its origin, and the blanked cells that are on the Task 1 target list must still be filled; onset queues appear exactly at T+30; window origins follow published selection rules. Leaderboard feedback is coarse: one Task 2 window is worth 0.00375 of the total score and each condition has only 40 windows per month.

Approach. We organised the pipeline (Figure 1) around four principles: (i) use signals that survive the shift (flow at the origin relative to bottleneck capacity, the observation row at T, corridor topology, conservation); (ii) adapt to the new scenario only with data the rules allow (Task 1 is offline, so March and April visible cells may be used; Task 2 is causal, so April models are fine-tuned on March); (iii) encode documented evaluation structure

instead of fighting it; (iv) validate every change with proxies whose bias against the leaderboard we measured, and freeze decision rules before looking at the outcome wherever possible.

Contributions.

- A shift-aware Task 1 chain: a LightGBM residual model and a whole-day 2-D U-Net trained with random 90-minute corridor gaps, self-supervised month adaptation, and a sequence of bounded, month-specific calibration stages (per-link bias, restacking, blackout specialists, congested-cell L1 density refill, corridor-wide speed common mode, residual boosters) whose successive gains shrank to the leaderboard's 10^{-5} resolution.
- Task 2 onset recast as a choice between recurrent bottleneck blocks; on the two D12_I5 corridors the final rule compares each block's flow at T with its empirical breakdown flow and the recent demand step, and we show why this comparison is invariant to corridor-wide shifts and why the switch is instantaneous at 5-minute resolution (time-aware per-link models remain on three corridors where flow-based choosers fail).
- Task 2 ongoing ensembles (LightGBM, CatBoost, corridor 1-D U-Net, GRU+BiLSTM) with an expected-IoU decision, retraining for the origin row, April-only fine-tuning on March, and Bayesian conditioning on the official selection event; a per-class analysis shows that newborn queues, 22% of the ongoing windows, carry the largest share of the remaining ongoing loss (about 43%) and about two thirds of the fine-tuning gain.
- Two Task 2 queue guards applied after the model chain, without retraining (Section VI-E): a spatio-temporal guard that, as described by the team member who built it, gates corrections with neighbouring historical observations (94 cells), and a history guard that, in constrained ongoing windows, turns a cell to queued only where the 30 nearest training-period analogues of the window's history up to T give enough evidence (10 cells). Together they change 104 of the 174,000 queue cells of V84.
- An analysis of Tasks 3 and 4: Task 4 (implied $S_{\mathrm{ODME}} \approx 0.994$) has at most about 0.0012 of total score left, and its counts identify only the cross-section flows of each corridor; Task 3 (implied $S_{\mathrm{physics}} \approx 0.73$–$0.76$ against 0.964 for an exact answer) costs about 0.03 of total score, a gap we attribute mainly to unpredictable accumulation noise at hidden cells, and we show which Task 1 changes still move it.
- A measured account of proxy-versus-leaderboard transfer, a cross-task error budget in total-score units, and complete compliance disclosures.
- A single-entry reproduction package: a clean copy holding only the official data and our package (code, trained models and frozen decision files) rebuilt both final files with content identical to the submitted ones (18.2 h of measured stage time on one laptop); the Task 2 queue files are rebuilt from the shipped models byte-identically, and all 160 official windows of V84 are re-derived with zero mismatching cells; the history guard is recomputed from frozen member probabilities, while the spatio-temporal guard, whose code is not part of the package, enters as a frozen list of its 94 flips.

Results. The model chain ends with V84, which scores 0.90336 on the public leaderboard (attributed from the change of the team best, Table VIII); Figure 2 shows how it was reached. The two selected final files are V84 plus Task 2 queue guards (Section VI-E): V86 flips 94 queue cells by the spatio-temporal guard, and V87 flips the same 94 cells plus the 10 cells of the history guard, which was also submitted on its own as V85 (public 0.90345, not selected). V86 and V87 score 0.90359 and 0.90368 on the public leaderboard; on the private leaderboard V86, the better of the two, scores 0.89962 (V87: 0.89956), 7th of 802 teams on the Kaggle private leaderboard as displayed on 9 November 2026 (the organisers set the final standings from eligible submissions). Their Task 1 and Task 4 rows (and with them the inputs of the Task 3 score) are byte-identical to those of V84, so everything we report on these tasks holds for the final files. A registered hedge on V84 was built but not selected (Section X).

Outline and naming. Section II states the benchmark and rules, Section III the empirical findings that drive the design, Section IV the validation methodology, Sections V to VII the methods per task, Section VIII the error budget, Section IX what failed, Section X compliance and disclosures, and Section XI reproducibility. Version tags VN are our numbered builds. Several Task 1 stages were built as separate pieces and first scored inside the next numbered file; we then give the scored file in brackets, e.g. V57blk [V58]. The appendix lists every scored file with a logged score (Tables VIII and IX) and gives a glossary of the terms we use (Table XII).

II. Benchmark, Evaluation and Rules

A. Data

The release covers 10 directional corridors ("panels") in 5 families: D7_I10, D7_I210, D7_I405, D12_I5 and D12_I405, i.e. Caltrans district and freeway, each in both directions (suffix N/S/E/W). Every mainline link carries one detector: 44 to 352 links per corridor, 1,583 in total, plus 972 ramp detectors. Links are indexed by milepost order; we call this index the link *position*, and traffic runs towards higher positions on N/E corridors and towards lower positions on S/W corridors. Time runs in 5-minute steps, 288 per day. Train (2030-06-01 to 2031-02-28, 273 days) ships the unmasked layer next to the masked view; validation (March 2031, 31 days, 5 incidents) feeds the public leaderboard; private (April 2031, 30 days, 7 incidents) decides the ranking. A cell is score-eligible when at least 75% of its raw samples were observed; this holds for 0.764 of cells in every corridor and split. Each day falls under one masking regime that hides 20% (R1), 30% (R2) or 50% (R3) of its eligible cells; these hidden eligible cells

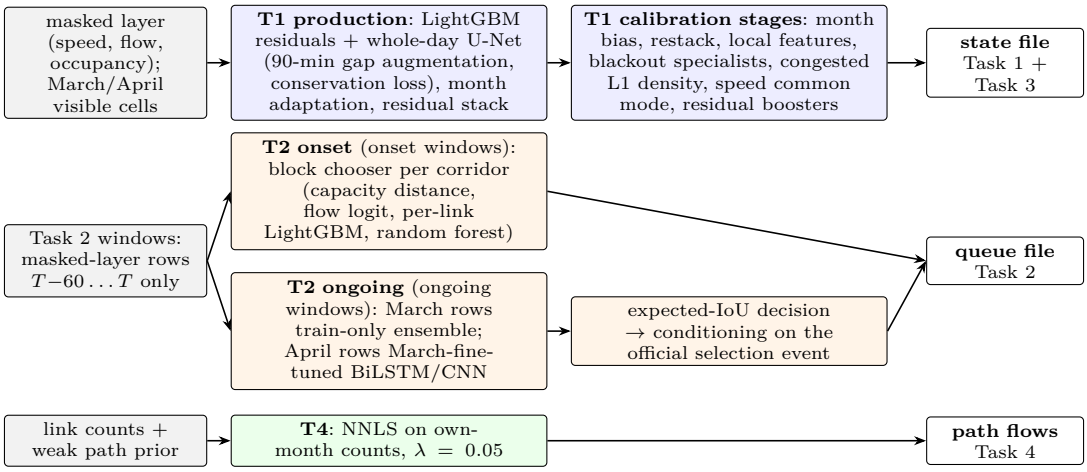


Fig. 1. Model chain up to V84, the base of the final files. Blue: Task 1 (and through it Task 3); orange: Task 2, where every input is a masked-layer row stamped at or before the window origin T; green: Task 4. The queue guards that turn V84 into V86 and V87 act on the queue file afterwards (Section VI-E) and are not shown.

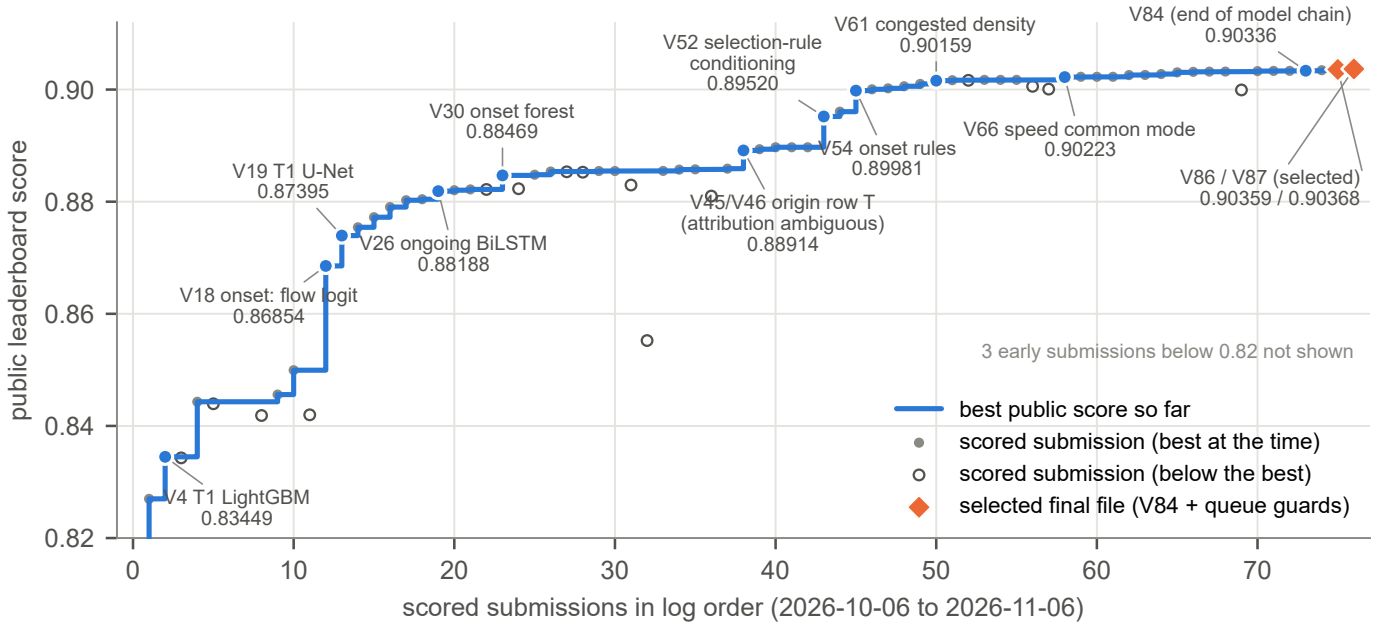


Fig. 2. Public-leaderboard score of every scored submission in our log, in log order (grey: best at the time; hollow: below the best), with the best-so-far curve and the main milestones. Orange diamonds: the selected final files V86 and V87, which are V84 plus the queue guards of Section VI-E; the grey point between V84 and them is V85, submitted alone. V45 is left out because its logged score duplicates that of V46; the jump marked V45/V46 cannot be attributed between them. The public board scores the validation month only; April-only changes (Section VI-C) leave it unchanged.

are the Task 1 targets, 3,451,376 in March and 3,284,419 in April. Congestion is rare: only 4.76% of eligible train cells are queued (observed speed at most $0.6\,v_f$).

B. Scores

Table I summarises the four task scores. Task 1 scores each regime as

$$S_{\text{state}} = 0.54\max\!\left(0, 1 - \tfrac{\text{RMSE}_v}{25}\right) + 0.46\max\!\left(0, 1 - \tfrac{\text{RMSE}_{q,\text{lane}}}{600}\right), \quad (1)$$

with speed in km/h and flow per lane in veh/h, averaged over the three regimes, then corridors, then families. The Task 1 answers are the unmasked observations of the training layer's kind, so they contain measurement noise. Task 2 covers 8 corridors (the D12_I405 pair is excluded) with 5 onset and 5 ongoing windows per corridor and split: in an onset window no link is queued anywhere in the history and at least one link is queued in the horizon; in an ongoing window at least one link is queued at T. Each window gives 60 minutes of history $(T-60, \ldots, T-5)$ plus, by organiser ruling, the observation row at the origin T; the forecast is

TABLE I
SCORING COMPONENTS (TOTAL $= 0.35S_{\text{STATE}} + 0.30S_{\text{QUEUE}} + 0.15S_{\text{PHYSICS}} + 0.20S_{\text{ODME}}$).

Task	Weight	Unit score and aggregation
1 state	0.35	Equation (1) per regime; mean over R1/R2/R3, corridors, families
2 queue	0.30	space-time IoU per window; condition → corridor → family; 8 corridors
3 physics	0.15	$\frac{1}{3}S_{\text{FD}} + \frac{2}{3}S_{\text{LWR}}$ on the Task 1 frame; exact answer 0.964 per the organisers' table
4 ODME	0.20	$0.45S_{\text{od}} + 0.25S_{\text{link}} + 0.15S_{\text{dev}} + 0.15S_{\text{attr}}$ per corridor

the queue state of every link at $T{+}5, \ldots, T{+}30$, where a link is queued when its *noise-free* speed is at most $0.6\,v_f$. A window is scored by space-time IoU; windows are averaged by condition, corridor and family, so one window is worth 1/80 of S_{queue} or 0.00375 of the total. Task 3 is computed from the Task 1 answer (our values at targets, released values at all other eligible cells): $S_{\text{physics}} = \frac{1}{3}S_{\text{FD}} + \frac{2}{3}S_{\text{LWR}}$, where S_{FD} is one minus the normalised L1 distance of per-lane flow to a triangular fundamental diagram [3] and

$$S_{\text{LWR}} = \max\Big(0,\, 1 - \frac{\sum |\Delta N - R|}{\sum |R|}\Big), \qquad R = \Delta t\,(q_{\text{in}} + r_{\text{on}} - q_{\text{out}} - r_{\text{off}}), \tag{2}$$

where $N = (q/v)L$ is the number of vehicles on a link of length L, ΔN its change over a 5-minute step Δt, $q_{\text{in}}, q_{\text{out}}$ the mainline boundary flows (held by the organisers) and $r_{\text{on}}, r_{\text{off}}$ the ramp flows [4], [5]. The scorer's documentation states that reproducing the published observations satisfies conservation exactly, while the organisers' baseline table lists $S_{\text{physics}} = 0.964$ for an exact answer and calls about 0.95–0.96 the practical ceiling for everyone; we use 0.964. Task 4 asks for flows on 35,354 paths, $S_{\text{ODME}} = 0.45S_{\text{od}} + 0.25S_{\text{link}} + 0.15S_{\text{dev}} + 0.15S_{\text{attr}}$: S_{od} compares path flows with a withheld reference, S_{link} the fit to the released link counts, S_{dev} the distance moved away from the released prior relative to the reference, and S_{attr} the destination-attraction distribution.

C. Task 2 window mechanics

Official windows follow published rules: origins at least 360 minutes apart, history coverage at least 0.7, a queue somewhere in the horizon, and for ongoing windows a persistence IoU (last history step repeated over the horizon) of at most 0.9. Because the selector takes the earliest qualifying origins, all official windows fall early in their month (March days 1–12, April days 1–17, median day 3). In onset windows the queue appears only at $T{+}30$; the organisers confirmed this, and in all 40 official train onset windows every positive cell sits at $T{+}30$. After each scored origin the whole corridor is blanked for $(T, T{+}90]$ in the validation and private layers. The blanking itself creates no Task 1 targets, but the targets that the day's masking regime hides inside these spans stay on the target list and are scored: 64,720 in March and 63,745 in April, about 1.9% of all targets, and a change confined to them moved the public score by +0.00037 (Section V-B). The training months contain no comparable gaps. Figure 3 shows one corridor-day.

D. Rules that shape the method

Tasks 1, 3 and 4 are offline: any observation of the same month may be used. Task 2 is causal. In forum rulings of 15 October the organisers stated that the origin row T may be used, that everything used for a window (inputs, fitting, model and threshold selection) must be stamped at or before its T, and that April windows may therefore use train plus the whole validation month, while April data may not be used for selection. At award review the organisers re-run each team's code and require the private score of the reproduced predictions to be within 1% of that of the selected final submission; for Task 2 they also delete every row after each window's origin and check that predictions do not change. Both rules shape the method: the Task 2 code of our model chain and of the V85 guard reads only history up to T, and every decision is frozen in code or in shipped configuration files. For the spatio-temporal guard of V86 and V87, whose code is not in the package, the decisions ship as a frozen list of flipped cells, and we cannot show from code that it uses only data up to T (Section X).

III. What the Synthetic Traffic Taught Us

Most design decisions follow from a handful of empirical regularities. We list them first because they explain why the methods in Sections V to VII look the way they do.

A. Queues are bottleneck blocks that switch on at breakdown

Congestion in the generator forms at a small set of recurrent bottlenecks. An onset is a whole block of adjacent links dropping to congested speed in one 5-minute step, with no slowdown before it: on D7_I10_E, positions 4–10 fall together from about 113 to 37 km/h. This is what kinematic-wave theory predicts at 5-minute resolution [4], [5]. With the released triangular fundamental diagrams (jam density 100 veh/km/lane; critical density 17–18 on the eight Task 2 corridors), the backward wave speed $w = C/(k_{\text{jam}} - k_c)$ has corridor medians of 23–26 km/h on those corridors (23–28 over all ten), about 2 km per step, while the recurrent blocks are 0.1–2.5 km long; the jam front crosses a block within about one step and leaves no precursor in the history. Five of the ten recurrent blocks on the four multi-block corridors sit immediately upstream of a link whose nominal capacity is 11–49% below the smallest capacity inside the block (on D12_I5_N, block 103–105 has 6 lanes and 12,500–12,600 veh/h and is followed by a 3-lane link of 6,314 veh/h); the others show no nominal reduction, so nominal capacity alone does not identify them, and our choosers use the empirical breakdown flow

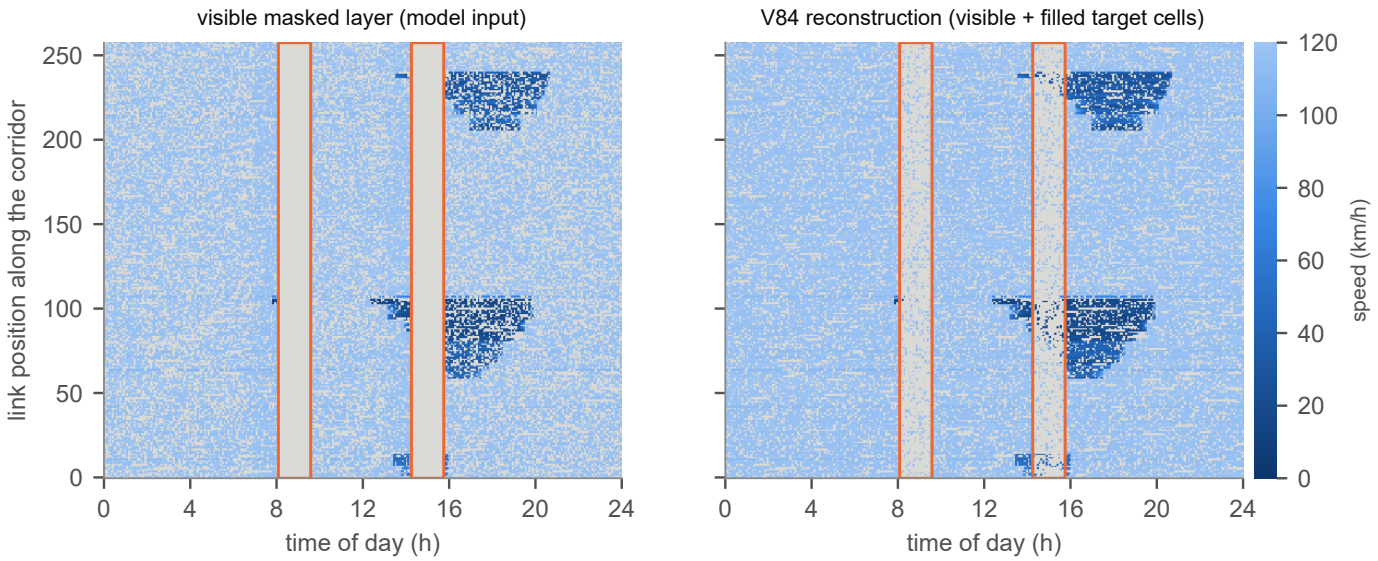


Fig. 3. One validation corridor-day (D12_I5_N, 12 March 2031). Left: the visible masked layer the models see. Right: the V84 state (Task 1 rows identical in the final files V86 and V87), with visible cells kept and Task 1 target cells filled by the submission. Orange boxes: the 90-minute blackouts after the two official ongoing origins of that day (08:00 and 14:10); all links are hidden there, and the regime-masked targets inside them are filled from observations up to 45 minutes away. Grey: hidden cells that are not Task 1 targets (unscored). Two congestion fields form in the early afternoon and dissolve by about 20:00; their heads sit at the recurrent blocks 103–105 and 236–238 (Table IV), just upstream of capacity drops, and they grow upstream (towards lower positions), as LWR predicts for a bottleneck queue. The second blackout cuts into their growth phase.

of each block instead. Demand moves in steps of roughly 30 minutes, and the block that breaks tends to be one whose flow at T is close to its breakdown flow; a chooser built on this closes part, though not all, of the gap to an oracle (Section VI-A). Ongoing queues grow and dissolve along kinematic waves, so the information about an ongoing queue's future sits in its current extent, its head and tail, and the flows feeding it.

B. The scored months are a different realisation

Validation and private come from one scenario with an organiser-held random seed (the release contract lists a capacity multiplier of 1.2 and incident severity 1.0 without stating the training values). The visible consequence is that different blocks break first (Figure 4). On D12_I5_S, block 43–45 is the first to break in 62% of training pseudo onset windows and in none of the 26 March pseudo onset windows; on D7_I405_S block 32–39 rises from 50% to 89%. Validating by leaving one training month out cannot detect this, because all nine training months share one realisation of the scenario.

The shift also appears in Task 1. Per-link residual biases of the production model correlate at 0.97–0.99 between odd and even days of a month but only at 0.36 across months, and predictions are over-dispersed inside free flow. Month-specific recalibration from each month's own visible cells was the largest single Task 1 stage (Section V-B).

What transfers and what does not. Train-to-March transfer separates two kinds of knowledge. *Location and timing priors* (which block breaks first, at which time of day) belong to one demand realisation; on the multi-block corridors, onset choosers that learned them reversed on March, and on D12_I5_S April's own data before the window origins contradict March's onset habit (Section VI-A). *Queue dynamics* (how fast a queue of a given extent and inflow grows or dissolves) depend on the scenario's capacities, which March and April share. Fine-tuning the ongoing models on March raises cross-fitted March IoU of newborn queues by 0.071 but of mature queues by only 0.011 (Section VI-C). An adaptation of growth dynamics would produce this, but so would March-specific knowledge of which bottleneck is active, which both half months share; March data cannot separate the two. We therefore use March for April ongoing rows but not for April onset locations, and expect part of the cross-fitted gain not to carry over to April.

C. Truth definitions differ between tasks

Task 2 labels use the noise-free (latent) speed, whereas the Task 1 answers are the released noisy observations. The difference is visible in our results. About 32% of the free-flow speed noise variance is a corridor-wide shock shared by all links at a time step; adding this shock, estimated from the other visible links, to the free-flow predictions (V66) realised slightly more than its offline estimate on the public board (+0.00050). For Task 2 the opposite holds: a lane of ongoing members fed the noisier non-eligible observations passed its pre-registered offline test but lost 0.00165. Task 1 rewards modelling measurement noise that is shared across links; Task 2 rewards ignoring it. All our Task 2 models are nevertheless

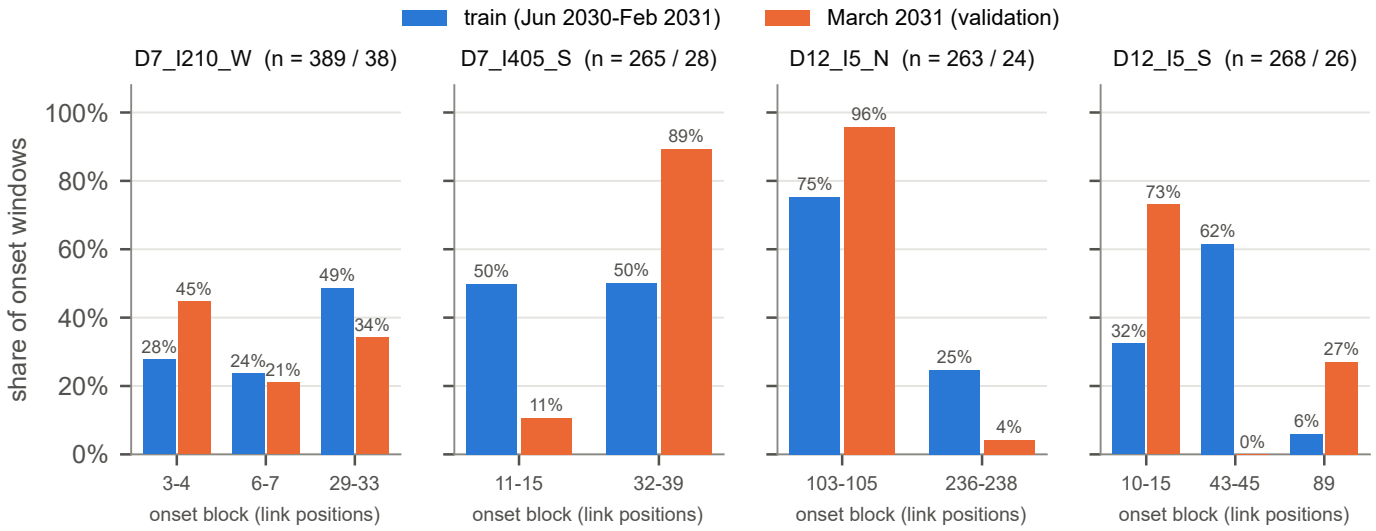


Fig. 4. Which bottleneck block breaks first, train months versus March (validation), for the four Task 2 corridors with several recurrent blocks (labels: link positions). Shares are over onset pseudo windows with a defined block target (n = train / March). The March habit differs sharply from train on D7_I405_S and D12_I5_S. April is not shown; the uses of April data for choices (pseudo windows until V45 and two later decisions that looked at April visible data, one of them for the hedge, which was not selected) are disclosed in Section X.

trained and evaluated on labels from observed speed (the only labels available), so cells near the queue threshold are noisily labelled throughout our offline work.

D. Blackouts are where Task 1 loses most

The official horizons blank all links for 90 minutes, so a target inside a blackout can be up to 45 minutes from the nearest observation on any link. Nearest-reading features become meaningless there; on training data with simulated 90-minute gaps the production LightGBM had a speed RMSE of 11–14 km/h inside the gap against about 1.9 outside (from our notes; that run saved no result file). The production fill used until V57 (two month-adapted U-Nets) misses by 4.7–10.7 km/h RMSE inside simulated official blackouts versus 1.3–2.2 outside (Figure 5); queued cells are about 8% of blackout cells but carry 53–55% of the blackout speed error, and ongoing windows hold 59–62% of the blackout cells but 73–75% of the error. Because Task 1 is offline, the states before and after a gap are both visible; what is missing is *when* a queue dissolved or grew inside it.

E. Task 3 measures accumulation against noisy data

The organisers' boundary flows are projected onto the released observations. S_{LWR} is then effectively the L1 error of our accumulation changes at target cells, normalised by a signal that is only about 0.75% of N. Because the answers contain measurement noise, ΔN at a target cell carries noise that no reconstruction can predict from its neighbours. The score rewards getting *density* right at congested cells (where $k = q/v$ is large and sensitive), not speed or flow in isolation; of all Task 1 stages, only the congested L1 density refill (V61cg) and the corridor-wide speed common mode (V66) realised their conservation-heavy estimates in full (Section VII).

IV. Validation Methodology

A. What the leaderboard can and cannot tell

The public board scores one month. For Task 2 it averages 40 windows per condition; with the organisers' per-window IoU standard deviation of 0.375, a condition's level has a standard error of about 0.059 IoU, i.e. 0.009 of the total score. Two variants scored on the same windows differ only where a change touches them: a full flip of one window is worth 0.00375; our onset changes altered one to three March windows, while ongoing changes touched 10–27 March windows by a few cells each (V28: 12 cells in 10 windows; V48: 56 in 20; V52: 102 in 20). A public Task 2 delta is therefore an exact measurement on March's windows but a weak predictor of the April effect. Task 1 stages are different: they rewrite millions of state rows and leave queue and Task 4 rows byte-identical, so their public deltas are close to deterministic measurements of the March effect, at the five-decimal resolution of the board. We used Task 1 deltas to calibrate our offline proxies; Task 2 public deltas were used to reject changes that lost (V12, V43, V67dg, V80) and, through single-window probes, to learn which March onset blocks were right; no probe edit is part of the final files V86 and V87 (Section X). The public scores of the guard files V85–V87 were known when the final pair was selected (Section VI-E); like other Task 2 deltas, they measure the effect of the changed March cells (62 in V86, 70 in V87) exactly and say little about the April cells (32 and 34).

B. Task 2: pseudo windows and the replicated selector

Pseudo windows (`t2_pseudo.py`) cut onset- and ongoing-like windows from each month's visible masked layer (about 3,440 ongoing windows per month), with labels from visible speeds (noisy, and only on visible cells). They rank ongoing variants reasonably but overstate absolute IoU (V48: 0.9014 pseudo versus about 0.84 implied by the leaderboard), partly because they are drawn from the whole month while official windows are each month's first events. Realised leaderboard gains of adopted ongoing steps were about 0.6 of the pseudo-implied gains, with large scatter between steps, and pseudo-positive variants that were rejected on the board would lower the ratio further. Applied to June 2030, the pseudo rules reproduce none of the 40 official train windows. **Replica windows** implement the official selector instead (shared 360-minute spacing across conditions, earliest qualifying origin, persistence computed on the latent queue); they match 39 of 40 official June onset windows and 32 of 40 ongoing ones. Applied to March, replica windows are the only out-of-scenario ruler for Task 2 rules.

C. Task 1: hide frames, rulers and held-out checks

Hide-20% frames. For each scored month, 20% of every day's visible eligible cells are hidden again (fixed seed) and the full production chain is re-run on them, which gives labelled pseudo targets from the right month (1.44M / 1.41M cells for March / April). Configurations are chosen by odd/even-day cross-fitting with a pre-declared objective $0.35\,\Delta S_{\text{state}} + 0.15\,\Delta S_{\text{physics}}$, where the physics part uses an LWR proxy that replaces the organisers' boundary flows by the true accumulation change (Section VII); nested month checks, λ curves, reweighting of hidden cells to look like real targets and a real-frame S_{FD} gate follow. **The blackout ruler** simulates 977 official-like blackouts (494 in March, 483 in April; its March candidate origins are the replica origins, minus those whose span touches an official blackout) and scores gap fills against the visible observations, with block halves and a window bootstrap. **Six held-out checks** were required for every late Task 1 version (V73–V84): parity days, half months and the opposite month, in both March and April; from V81 on the amplitude was chosen on parity and then frozen, and each check additionally needed a lower LWR proxy and at least 7 of 10 corridors improving. Hidden-value poison tests (overwriting all hidden channels and asserting unchanged features) guarded against leakage.

D. Task 2 April models: half-month cross-fitting

Models used only for April windows are fine-tuned on March. To choose them on March without scoring a model on days it was trained on, every fine-tune is run twice, on days 1–14 (scored on days 15–31) and on days 15–31 (scored on days 1–14); the April model is the fine-tune on all of March. Decisions on April models were taken only on these cross-fitted March numbers. The limits of this protocol are discussed in Section VI-C.

TABLE II
OFFLINE ESTIMATE VERSUS REALISED PUBLIC GAIN (TOTAL SCORE); OTHER TASK 1 STAGES ARE IN TABLE III. TASK 1 ESTIMATES USE THE FULL LWR PROXY, EXCEPT THE V73–V84 ROW, WHICH SUMS THE CONSERVATIVE PER-STEP PROXIES (MINIMUM OVER THE SIX HELD-OUT CHECKS, REAL-FRAME S_{FD}, LWR PART COUNTED AT 40%).

Build	Change	Estimate	Realised
V61cg	T1 congested L1 density	+0.00060	+0.00058
V63	T1 density refill in blackouts	+0.00005	−0.00002
V73–V84	T1 residual boosters (11)	+0.00080	+0.00108
V52	T2 selection-rule conditioning	+0.0014	+0.00549
V67dg	T2 degraded-observation members	positive	−0.00165
V80	T2 trajectory CNN (train gate)	positive	−0.00328
V43	T2 onset consensus switch	positive	−0.00477

E. Proxy versus leaderboard

Table II contrasts offline estimates with realised public gains for representative changes (the remaining Task 1 stages are in Table III). Two patterns stand out. First, the state part of Task 1 proxies transfers almost fully, while the LWR part transfers unevenly: about 40% for most free-flow stages, about 100% for the first congested-density refill and for the speed common mode, nothing for a congested refill inside blackouts. After V64 we counted the LWR part at 40% when estimating gains. Second, the Task 2 changes that rested only on pseudo, replica or training-month evidence without encoding the official window structure were unreliable: the three failed Task 2 changes in Table II are of this kind, while the one that encoded the selection rule realised far more than predicted.

V. Task 1: State Reconstruction

The final state file (the Task 1 rows of V84, identical in V86 and V87) is the output of a production chain (Section V-A) followed by bounded calibration stages that each rewrite one disjoint class of rows: free-flow targets outside blackouts (about 6.34M rows), congested targets outside blackouts (269,474 rows) and targets inside official blackouts (128,465 rows).

A. Production chain

LightGBM residual model [6]. The baseline interpolates speed in time and blends temporal and spatial estimates for flow. For each target, 46 features describe the same day around it: nearest readings before and after with their gaps, nearest upstream and downstream readings in travel order with distances, neighbours' temporal estimates, weekday and time-of-day profiles from visible training data only, deviations from those profiles, observation coverage, ramps, geometry and regime. Three regressors predict residuals of speed, per-lane flow and length-weighted L1 density, and their outputs are reconciled in log space with equal weights; folds alternate by

month parity. Out of fold on all ten corridors the speed and flow regressors lift S_{state} from 0.9148 (interpolation) to 0.9370; on three corridors the density reconciliation then trades about 0.002 of S_{state} (0.9319 to 0.9295) for about 0.03 of physics proxy (0.662 to 0.692).

Whole-day U-Net [7]. A 4-level residual 2-D U-Net (GroupNorm, GELU) sees a whole day as an image of 288 time steps by links in travel order with 23 input channels (observations and visibility, coverage, profiles, the interpolation baseline, static link features, time of day, weekend and regime), and predicts residuals of v/v_f and q/cap (free speed and capacity of the link). The loss is squared error in score units (0.54 speed/25, 0.46 flow/600) on targets and extra hidden visible cells. Two augmentations matter: 60% of training days receive one or two corridor-wide 90-minute gaps, imitating the official blackouts, and 8% of cells are hidden at random. A variant adds $0.05\times$ a conservation term $\sum|\Delta N_{\text{pred}} - \Delta N_{\text{true}}| / \sum|\Delta N_{\text{true}}|$ over 5-minute transitions, which ties the network to the Task 3 objective. The width-48 network has 7.29M parameters and trains 30 epochs per fold (about 30–40 minutes on an RTX 4070 Laptop GPU); the scored months use the mean of the two fold models. The blend gives the U-Nets all weight inside blackouts, where the LightGBM has no valid neighbours. On one fold of four corridors, with blend weights chosen on the same out-of-fold predictions (hence optimistic), the objective $0.35S_{\text{state}} + 0.15S_{\text{physics}}$ (physics part by the LWR proxy of Section VII) rose from 0.4294 (LightGBM) through 0.4344 (first U-Net blend) to 0.4393 (two width-48 nets, one with the conservation term). The first U-Net blend added 0.00541 on the public board, close to its offline gain of 0.0050.

Self-supervised month adaptation. Both nets are fine-tuned on each scored month's own visible cells: random visible cells and 90-minute gaps are hidden and the hidden observations serve as labels (production: learning rate 10^{-4}, 6 epochs, both nets and folds). In a self-check (one fold-0 net adapted for 3 epochs on even days, scored on hidden visible cells of odd days) speed RMSE fell from 1.977 to 1.871 km/h with flow RMSE unchanged (32.53 vs 32.58 veh/h); similar short adaptations on held-out training months were neutral or slightly negative (objective 0.4386 to 0.4368/0.4385), consistent with a shift that exists only between training and scored months. This is legal for Task 1 because it is offline; the corresponding Task 2 move would use rows after many windows' origins and is forbidden (Section VI-C).

Residual stack. A small LightGBM fitted on out-of-fold ensemble outputs of three training months predicts residuals of speed, flow and density and projects the result onto the corrected density (V30/V31); it was transferred to more corridors under a pre-registered gate (V36) and halved where its effect was ambiguous (V40). Together these steps added only +0.00023 on the public board (V31, the stack on the V28 queue rows, +0.00012 against V28, confirmed as V32 on the V30 queue rows, +0.00012; V36 +0.00009; V40 +0.00002), an early sign that the train-fitted chain was close to its limit and that the remaining error was month-specific.

B. Calibration stages

Table III lists the stages that reached V84 (and hence the final files); the hide frames exposed three defects of the production chain that they correct.

Month-specific miscalibration. Per-link biases are stable within a month but not across months, and the chain is over-dispersed inside free flow. V52t1 [V53] shrinks per-link, per-month speed and flow biases estimated on predicted-free hide cells; V56 refines them by speed band and time of day; V57rs [V57] re-stacks the chain's components with month-specific LightGBM weights; V59loc [V60] adds local neighbourhood features.

Blackouts. V57blk [V58] fits a residual LightGBM on the production gap fill using the blackout ruler; V59blk [V60] adds gap-specialist U-Nets fine-tuned on official-like gaps in training months plus a residual model; E1 [V62] averages the two specialist fills. On the ruler the three rounds recovered about 0.0008 of the roughly 0.004 that blackouts cost against a perfect fill (Section VIII).

Congested density and shared noise. Congested cells are 3.7–4.0% of hide cells but carry 20–22% of the density L1 error, and for such a cell the conservation term weighs about ten times its speed term. V61cg [V61] refills congested non-blackout targets with a LightGBM trained on an L1 loss in density space (an L2 mean shift lowered speed error but hurt conservation, because $N \propto 1/v$ at congested cells); V64 refits it on both months. V66 adds to each free-flow target the corridor-wide, per-time-step speed shock estimated by least squares from the other visible links (first differences across the corridor and ±15-minute deviations, fitted per month on hide cells), so that predictions follow shocks that are part of the scored observation; V69 adds a second-stage LightGBM on its residual.

C. Residual boosters V73–V84

After V69, eleven bounded residual models, each fitted only on visible observations and each passing the six held-out checks of Section IV, removed three remaining kinds of structure from non-blackout targets (lineage in Table X). *Day-level per-link speed offsets*: additive monthly and daily factors with temporal smoothers clipped to ±1 km/h, low-rank factors and a LightGBM mixer of both (V73–V75, together +0.00049 public). *Local spatio-temporal context*: a speed mixer of neighbouring speed, flow, interpolation and slope features within ±2 km/h of the V69 anchor, plus an independent flow residual (V76, +0.00030). *Flow detail*: a 77-feature flow model capped at 60 veh/h per lane, grown in continuation steps to 1,600 trees, and an observation-only 180-tree residual with 43 multiscale speed, flow, density, occupancy and coverage features (V77–V79, V83, V84, together +0.00017). Congested speed received two common-mode ridges within ±1 km/h of V79 (V81, V82,

TABLE III

TASK 1 CALIBRATION STAGES IN THE FINAL LINEAGE, LABELLED BY BUILD AND, IN BRACKETS, THE SCORED FILE THAT FIRST CONTAINED THEM (PUBLIC = VALIDATION MONTH). ESTIMATES USE THE HIDE-20% FRAME OR THE BLACKOUT RULER; "LOWER BOUND" COUNTS ONLY S_{STATE} AND THE REAL-FRAME S_{FD}.

Build [scored as]	Rows changed	Method	Estimate	Lower bound	Public gain
V52t1 [V53]	6.34M free	per-month per-link speed/flow bias on predicted-free cells	+0.00101	+0.00074	+0.00086
V56	6.34M free	per-link × 7 speed bands, per-link flow scale, 4-hour time-of-day offsets	+0.00031	+0.00019	+0.00024
V57rs [V57]	6.34M free	month-specific LightGBM restack of component deviations	+0.00023	+0.00017	+0.00017
V57blk [V58]	128k blackout	residual LightGBM on the production gap fill	+0.00040	+0.00031	+0.00037
V59loc+blk [V60]	free + blackout	local-feature restack; gap-specialist U-Net + residual model	+0.00058	+0.00030	+0.00042
V61cg [V61]	269k congested	L1-density LightGBM refill	+0.00060	+0.00005	+0.00058
E1 [V62]	128k blackout	mean of two specialist gap fills	+0.00011	+0.00008	+0.00008
V64	269k congested	density model pooled over both months	+0.00015	≈0	+0.00006
V66	5.67M free speed	corridor-wide per-time speed shock added	+0.00049	+0.00036	+0.00050
V69	5.60M free speed	second-stage LightGBM on the common-mode residual	+0.00005[a]	+0.00004	+0.00005

[a] LWR part counted at 40%.

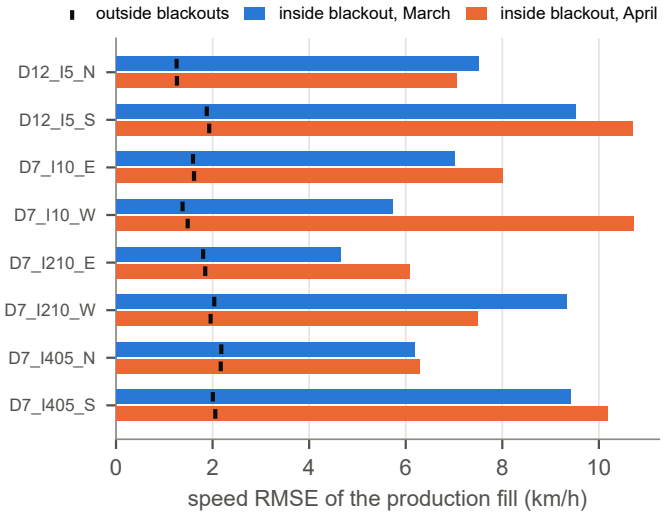


Fig. 5. Speed RMSE of the production gap fill (two month-adapted U-Nets, used until V57) inside simulated official-rule blackouts and outside them, per corridor, on the blackout ruler.

+0.00012). Gains decayed from $3 \cdot 10^{-4}$ for the first offset and context models to 10^{-5} per step, the leaderboard's resolution; the conservative proxies summed to +0.00080 against a realised +0.00108, and four versions (V73g, V75, V77, V84) improved all ten corridors in all six checks. Most values written by V73–V75 are later overwritten (179,860 V75 speed rows and 4 V73/V74 rows survive in V84 and hence in the final files), and their factors remain inputs of the V76 speed mixer and of the flow model. A queue-trajectory network built on V79 (V80) lost 0.00328 and is a dead branch; V81 was built on V79.

VI. TASK 2: QUEUE FORECASTING

A. Onset as a choice between bottleneck blocks

Because an onset is a whole block switching on at T+30 (Section III-A), we predict nothing at T+5, ..., T+25 and one link set at T+30. **Blocks** are extracted from training onset events: a link is kept if it is an onset link in at least max(3, 3%) of events; kept links within three positions that co-occur in at least 30% of the rarer link's events are merged; a block's canonical links are the members present in at least 30% of its events. **Choosers.** The flow model is a conditional logit over blocks with relative and absolute log block flow at the end of the history and its trend against the start of the history, deliberately without a time-of-day term. The capacity-distance model scores block b by $\theta_0(\log q_b - c_b) + \theta_1 \,\text{trend}_b + a_b$ with one slope shared by all blocks, c_b the median log flow of training onsets of b (its empirical breakdown flow) and a_b a ridge-penalised per-block intercept fitted on training onsets; sharing the slope keeps rare blocks such as 89 from being fitted to noise. The flow terms have coefficients shared by all blocks, so a corridor-wide multiplicative change of capacity or demand adds the same constant to every block's score and cancels in the softmax; the per-block constants $a_b - \theta_0 c_b$ are fitted on training months and do not react to such a change. The chooser therefore reads where the corridor is *relatively* closest to breakdown, which is what survives a change of scenario. V54 extends it with a 15-minute trend and the 30-minute demand jumps visible at T (for D12_I5_N also the origin row). On D12_I5_S the fitted coefficients are 72.5 on the capacity distance and −45.5 on the 15-minute trend (log flow over $T-15..T$ minus $T-30..T-15$), so the score is $27.0 \log q_{[T-15,T)} + 45.5 \log q_{[T-30,T-15)} - 72.5\, c_b + a_b$ with $a = (0, 1.15, 3.04)$ for blocks 10–15, 43–45 and 89, plus +16.2 for a demand step 35 minutes before T (smaller jump terms omitted). The block that breaks is the one

whose flow has been held near its breakdown level for half an hour after a demand step, not the one whose flow is still climbing: once a bottleneck reaches its discharge limit its throughput stops rising, and the excess demand is stored upstream until the queue appears.

Why habit fails. On the multi-block corridors, choosers that used time of day, demand phase or a time-aware per-link model learned the training habit and lost on March: a consensus block switch on D7_I405_S (−0.00477, both public windows wrong) and, on March replica windows, the literal pre-registered rule, which would have picked the per-link LightGBM habit on three corridors (−0.133, −0.190, −0.385 IoU). A month-aware prior built from each month's own earlier onsets also lost (−0.00209 against its base), which we attribute to noisy pseudo-window evidence. The exception is D7_I210_W: its early-morning block 6–7 breaks at low flow, where flow and capacity models collapse (training IoU 0.68 for the flow logit and 0.43–0.61 for capacity models without phase terms, against 0.96 for the ml2 block rule), so a time-aware per-link LightGBM with window weighting and strong regularisation ("ml2") is kept, and only its decision changed from a 0.4 threshold to the block with the highest mean probability.

Rule selection. V54 evaluated about 20 candidates per corridor on replica onset windows with a train leave-one-month-out rule fixed in advance; habit families were excluded after the March replica results showed their reversal, a March-informed deviation (Section X). With this many candidates the passing margins are optimistic: the D12_I5_N rule passed only weakly (lower bound +0.003), block definitions use all training months (a small leak shared by all candidates), and because March replicas shaped the candidate set, the +1 public window below is not an independent confirmation. Table IV lists the final rules. V54 changed three public windows and two April windows and gained exactly one window (+0.00375). Single-window probes on the public board later confirmed several of these blocks, but no probe edit entered a final submission: the March onset rows of V84, which the final queue guards (Section VI-E) leave unchanged, are those of V54, built before the probe rounds that tested them (an earlier probe file on one of the same windows is disclosed in Section X), and the organisers' rows-after-T re-run would expose any hard-coded window.

April onset. For April windows the whole of March may be used. March refits, a March time-of-day prior and a causal within-month update of the block prior agreed with production on every April window of D7_I405_S and D12_I5_N; on D12_I5_S, April's own visible data before the window origins show block 43–45 queueing in daytime, which March never showed. April onset therefore keeps the train-fitted rules; because this decision looked at April visible data, it is listed among the disclosures (Section X).

TABLE IV
FINAL ONSET RULE PER CORRIDOR (ONSET ROWS OF V84, UNCHANGED IN V86 AND V87). BLOCKS ARE GIVEN BY LINK POSITIONS.

Corridor (blocks)	Rule
D12_I5_S (10–15 / 43–45 / 89)	capacity distance + 15-min trend + 30-min demand jumps (V54)
D12_I5_N (103–105 / 236–238)	as above, with the origin row (V54)
D7_I405_S (11–15 / 32–39)	flow logit (V18; capacity model in V24–V53; re-adopted by V54, no official window changed)
D7_I210_W (3–4 / 6–7 / 29–33)	ml2 per-link LightGBM, best block by mean probability (V54)
D7_I10_W (single block)	flow logit over training blocks (V18)
D7_I405_N (set level)	random forest over complete training queue sets, 180 trees, seed 3030 (V30)
D7_I10_E, D7_I210_E	ml2 probability ≥ 0.5, else the most likely link

B. Ongoing members and decision

Members. (i) A per-(link, step) LightGBM on neighbourhood features (used until the T-anchored retraining); (ii) the regularised ml2 LightGBM; (iii) CatBoost [8]; (iv) a corridor 1-D U-Net over all links' 60-minute histories (three seeds; "CNN" in tables and figures); (v) a corridor recurrent network: a GRU over each link's history followed by a two-layer bidirectional LSTM along the corridor (0.75M parameters) [9], [10], trained with binary cross-entropy plus a soft-IoU term [11]. The BiLSTM is the strongest single member: queue heads and tails propagate along the corridor, which a sequence model running in both travel directions captures naturally. Replacing the hand-made persistence rules by the first LightGBM was the largest single ongoing gain on the public board (+0.00946 against its base); later member additions moved the board by +0.00018 to +0.00180 per step, increments within the noise of 40 windows.

Decision. Members are averaged with weights chosen on pseudo windows and smoothed 25% towards the mean of train-only co-queue groups (adjacent links whose training queue states have Jaccard similarity of at least 0.95; +0.00014 public). Each window is then turned into a binary map by sorting its cell probabilities $p_{(1)} \geq p_{(2)} \geq \dots$ and keeping the k cells that maximise $\sum_{i\leq k} p_{(i)} \big/ \big(k + \sum_i p_i - \sum_{i\leq k} p_{(i)}\big)$, the ratio of expected intersection to expected union (+0.00057 public). The rule has no threshold to tune, which matters when there are too few windows to tune one, but it assumes calibrated probabilities.

Origin row and T-anchored retraining. After the ruling that the row at T may be used, shifting the existing members' inputs by one row raised March pseudo ongoing IoU from 0.8823 to 0.8958; members retrained to forecast directly from a history ending at T ("T-anchored")

reached 0.9014. The public board rose to 0.88971 over these steps; the attribution of the first jump is ambiguous because the log records 0.88914 for two consecutive builds. The March ongoing rows of V84, the base of the final files, use this train-only ensemble (Table V).

C. April-only fine-tuning on March

For April windows the rules allow fitting on train plus all of March. We continue training the T-anchored BiLSTM and CNN members on batches that are half training windows and half March masked-layer windows (learning rate $3 \cdot 10^{-4}$ with a one-cycle schedule; March histories additionally hidden at a random rate of up to 20%) [12], [13]. A control that fine-tunes on training months' masked inputs with true labels gains nothing (+0.0003), and in training months the visible speed equals the released answer, so the March gain is adaptation to the scored months rather than to a label definition. Cross-fitted March ongoing IoU rose from 0.9014 (train-only blend) to 0.9197 with 24k fine-tuning windows, 0.9224 with more seeds, 0.9254 with 48k and 0.9276 with 96k windows, improving both half months at every step (Figure 6). On 283 March replica ongoing windows the gain is concentrated on *newborn* queues (formed shortly before T): +0.071 IoU, against +0.011 for mature queues whose $T-5$ origin was rejected only for persistence and +0.014 for windows bound by the 360-minute spacing. Each step had a decision rule fixed before training; the 48k step (V71) failed one direction of its nested half-month check (weights picked on days 15–31 scored 0.9322 on days 1–14 against 0.9330 for the previous blend) and was adopted anyway by team decision; the 96k step (V72) passed its rule. These numbers are optimistic in three ways that March data alone cannot remove: (i) blend weights were chosen on the same cross-fitted March scores (1,626 configurations for V50, 3,411 for V51), and the one nested check shows that weight-selection optimism can be as large as one step's gain; (ii) the two half months share March's per-link biases and bottleneck habits, so cross-fitting measures transfer within a month rather than from March to April; (iii) fine-tuning and evaluation labels are visible noisy speeds, while official labels use latent speed. These changes touch only April rows and are invisible on the public board; if the historical pseudo-to-leaderboard ratio of about 0.6 applied, the +0.0262 gain would be worth $0.0262 \times 0.15 \times 0.6 \approx 0.0024$ on the private score, an upper estimate rather than a forecast.

D. Conditioning on the official selection event

The selector takes the earliest qualifying origin. For an ongoing window that is not bound by the 360-minute spacing and whose history already qualified five minutes earlier, the only reason $T-5$ was not chosen is that its persistence IoU exceeded 0.9 while that of T does not; because the two horizons share $T+5, \ldots, T+25$, the queue must change mainly at $T+30$, the one step the earlier origin did not cover. The same earliest-origin logic explains

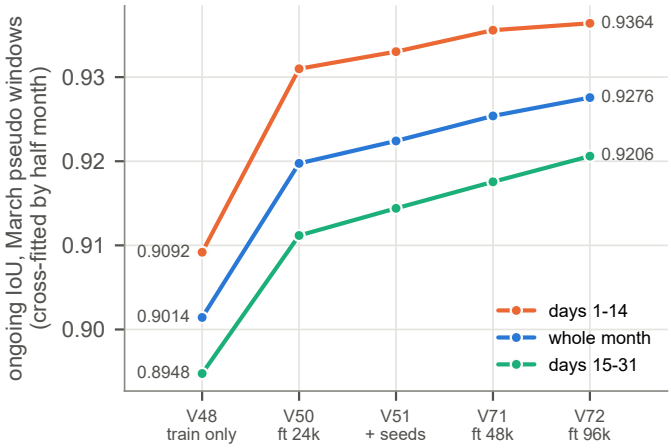


Fig. 6. Cross-fitted March ongoing IoU of the blends used for April rows: the train-only V48 blend and four March fine-tuning steps. Blend weights were selected on the plotted scores (not nested).

the onset rule: the earliest origin whose horizon contains a breakdown at time t is $t-30$. V52 exploits this "constrained" class (23 March and 20 April windows). It draws 20,000 joint samples from the ensemble's probabilities in which all links of a train-only co-queue group share one uniform variate (so marginals are preserved, the links of a group move together, and a link switches at most once when its probabilities are monotone over the horizon), approximates the latent state at $T-10..T$ by the sampled first horizon, keeps the samples consistent with "$T-5$ rejected, T accepted" (windows whose event probability is below 0.02 are left unchanged), and re-decides horizons 2–6 by the expected-IoU rule on the conditioned probabilities; evening windows on D7_I210_E use a dissolve rule instead. On March replica windows the rule gained +0.016 IoU per constrained window (day-clustered bootstrap 95% interval $[+0.0004, +0.0335]$), with most of the gain at $T+30$ (IoU 0.894 to 0.911, earlier steps within 0.005); on training months it gained +0.034 (9 of 9 months positive), but there the members are in-sample. The evidence is not uniform: on the 17 March replica windows most like the official ones (the first five per month) the gain was −0.001, it was negative on D12_I5_S, D7_I210_W and D7_I405_S (−0.0072, −0.0177, −0.0046), and the class premise holds only as far as our replica selector is exact (32 of 40 official June ongoing windows). With another Monte-Carlo seed 19–22 April cells would change, mostly in one window whose event probability (0.0188) sits at the 0.02 floor. On the public board the rule added +0.00549, about four times its out-of-sample prediction (+0.0014); on April's 20 constrained windows we expect about +0.0015 to +0.003. A truncation re-run that deletes every row after each origin changed none of the 87,000 queue cells.

E. Final queue guards

Each team selects two final submissions and, as usual on Kaggle, is ranked by the better of the two. Ours,

TABLE V
ONGOING RECIPE OF THE QUEUE ROWS OF V84; V86 AND V87 CHANGE 94 AND 104 OF THESE CELLS (SECTION VI-E).

Rows	Members (weight) → post-processing
March	T-anchored ml2 LightGBM (2), CNN 3 seeds (1), BiLSTM 2 seeds (5), origin-shifted older BiLSTMs (2); CatBoost mixed at 0.25 → group smoothing → expected-IoU decision → selection conditioning
April	March-fine-tuned BiLSTM seeds 0/1 at 96k windows (4), seeds 2/3 at 48k (2), CNN seeds 1/2 at 96k (4); no train-only member, no CatBoost → same chain

V87 and V86, are V84, the end of the model chain, with Task 2 `queue_pred` cells flipped by two guards that a team member ran on the queue rows of V84: V86 flips 94 cells (62 March, 32 April; 63 queued → not queued, 31 not queued → queued; 18 windows), and V87 is the union of these with the 10 cells of the V85 history guard (disjoint submission ids; 104 cells in 20 windows). All flipped cells lie in constrained ongoing windows (Section VI-D); every other byte, including all onset rows and every Task 1 and Task 4 row, equals that of V84. The guards re-decide single cells of V84; no model is retrained.

V85 history guard. It considers the 80 official ongoing windows but acts only on constrained windows, skipping the D7_I210_E evening windows, which use the dissolve rule, and windows whose event acceptance is below 2%. (i) *CPU replay:* from frozen CPU probabilities of the V84-era members (BiLSTM and CNN nets, ml2 LightGBM, CatBoost; 24 caches) it repeats the conditioning of Section VI-D with 20,000 group-coupled samples and seed `crc32(window_id)`; q_{cpu} is the mean of the accepted samples and the replay its expected-IoU decision. (ii) *Retrieval:* the query is the window's own history up to and including T (12 five-minute steps; at most 16 spatial bins per corridor, each with normalised speed, queue share and coverage; three successive 4-step means); among training-period origins (June 2030 to February 2031) within ±60 min of the query's clock time, the 30 nearest by mean squared feature distance (dates may repeat) give, for each cell of the next 5–30 minutes, the number of analogues with a valid observation, n_{valid}, and their queue share, q_{hist}. (iii) *Decision:* the window is decoded by the expected-IoU rule once from $0.9\,q_{\text{cpu}} + 0.1\,q_{\text{hist}}$ and once from q_{cpu}, the difference of the two decodings is applied to the replay with the first step kept fixed, and a cell is flipped only if this turns it from 0 to 1, V84 also has 0 there, $n_{\text{valid}} \geq 20$ and $q_{\text{hist}} - q_{\text{cpu}} \geq 0.30$. Of 12 candidates 10 pass the gate (8 March cells in 4 windows, 2 April cells in 2 windows); in the six modified windows the CPU baseline replay agrees with V84, so the flips do not come from differences between the replay and the submitted base. A window's query uses only the rows of its day up to and including T, the analogues and their futures come from the training period (the library is asserted to end before 1 March 2031), and, according to their generating scripts in the hand-off (shipped, not re-run), the cached member probabilities were computed from inputs ending at or before T, so the guard uses nothing after T of a scored day and satisfies the Task 2 rule. Stage `v85` recomputes the ten cells with the guard's own code (Section XI).

V86 spatio-temporal guard. We know its method only from the team member's hand-off: on top of V84, spatio-temporal linkage and gating by neighbouring historical observations correct queue predictions; a proposed flip whose CPU replay disagreed with the submitted base was not applied, and the formal build blocked four ambiguous flips, leaving 94. Its code is not in the package; its decisions ship as a frozen change list with each cell's value before and after (Section XI). Without its code we cannot show that it uses only data up to T; Section X discloses this.

Scores and local evidence. On the public board V84 scored 0.90336, V85 as a single file 0.90345 (not selected), V86 0.90359 and V87 0.90368, i.e. +0.00009, +0.00023 and +0.00032 over V84; these scores were known when the final pair was selected, and V87 was assembled after V85 and V86 had been submitted. Only the March cells (62 of V86, 70 of V87) reach the public board, and like other Task 2 deltas these measure the effect of the changed March cells exactly but say little about the April cells (32 and 34; Section IV). IoU is a nonlinear per-window score and four windows (three of them in March) hold cells of both guards, so the gains of the two guards need not add up. For the V85 rule the team member measured a proxy (labels from visible observations): on 251 March exploratory windows, panel-equal-weight proxy IoU rose by +0.000676 against the CPU replay baseline (day-clustered 95% interval $[-0.000324, +0.001747]$; 9 windows up, 4 down), the evidence gate added only +0.000027 over the ungated history rule ($[-0.000406, +0.000411]$), and 11 further March windows did not change. These windows had been used repeatedly to choose the rule and both intervals contain zero, so this is not an independent test (Section X); the hand-off gives no local numbers for V86. A registered private-only hedge, `v84_hedge.zip` (V84 with four April cells of the onset window `Q2_D12_I5_S_private_001` switched by a rule registered on 24 October), was built but not selected, and neither final file contains its switch.

VII. Tasks 3 and 4: Physics Consistency and Demand Recovery

A. Task 3

Task 3 has no rows of its own; it rescores the Task 1 frame. S_{FD} is dominated by the released cells: on the real submission frame after V61cg we measure 0.9881 (March) and 0.9880 (April), the level of an exact answer, so a Task 1 change moves it by about 10^{-4}. S_{LWR} cannot be scored locally: the public scorer's topology fallback gives zero even for the exact answer, because the organisers'

boundary flows are withheld. We therefore use a proxy that replaces the right-hand side of Equation (2) by the true accumulation change on hide cells, plus an empirical floor of 0.045. On training months out of fold the chain reaches S_{FD} 0.987, LWR proxy 0.595 and physics proxy 0.726.

The gap to a perfect answer. The organisers' table gives $S_{\mathrm{physics}} = 0.964$ for an exact answer, while our leaderboard decomposition (Section VII-C) implies about 0.73–0.76 for V84, whose Task 1 rows the final files share (assuming S_{state} between 0.94 and 0.95). The difference, about 0.03 of total score, is the largest single gap in our submission, larger than everything Task 1 stages gained after V48 (+0.0044) and both Task 2 rules (+0.0092) together. Almost all of it is in S_{LWR}. We attribute it mainly to measurement noise: the answers are noisy observations, ΔN at a target cell inherits that noise, and no reconstruction from neighbouring cells can predict it. A smoothed-truth oracle on training months reached only a physics proxy of 0.73–0.78 in our notes, against 0.714 for the chain of that time (0.726 for the final production recipe); this run was not logged, so we present the noise explanation as our best hypothesis rather than a measured bound. An earlier claim in our notes that Task 1 and Task 3 were "near their floor" (at V48) was later beaten threefold by the calibration stages, so we do not rule out further gains.

What moved Task 3. The conservation-loss U-Net is the first lever: in the four-corridor out-of-fold blend it lifted the physics proxy from 0.691 (LightGBM only) to 0.715, although a plain width-48 U-Net already reaches 0.712, so most of the lift comes from adding a whole-day network at all. The second is the congested L1 density refill (V61cg): an estimate that was about 90% LWR proxy realised almost fully (+0.00058 public), near the top of our registered expectation (about +0.0003, range +0.00005 to +0.0006). Free-flow stages realised their LWR part only partly (V52 44%, V56 42%, V57rs 8%), plausibly because the hide frame partly rewards fitting the noise of neighbouring observations; the speed common mode (V66) realised fully, so about 40% is a median, not a law. Density refills inside blackouts (V63), LWR-consistency smoothing of congested runs and L1 density on free-flow cells all failed.

B. Task 4

Our path flows are the output of the organisers' public builder (its "public reference" file, not the withheld scoring reference): non-negative least squares [14], [15]

$$\min_{f \geq 0} \ \|Af - c\|^2 + \lambda \|f - b\|^2, \qquad \lambda = 0.05, \qquad (3)$$

on each month's own released link counts c and weak prior b. It fits the counts almost perfectly (S_{link} 0.99994 / 0.99992), so a perfect link fit could add at most $4 \cdot 10^{-6}$. **What the counts can identify.** Every OD pair has exactly one path, so path flows are OD flows of a linear corridor: n zones give $n(n-1)/2$ forward pairs (630–5,050 per corridor). The measured rank of the count operator is exactly $n-1$ on every corridor (e.g. 101 zones: 5,050 paths, rank 100). On a linear corridor the counts therefore fix only the $n-1$ cross-section flows, i.e. the net balance of trips entering and leaving each zone, and say nothing about trip lengths or about productions and attractions separately; those come entirely from the prior and the regulariser. This explains three observations: NNLS sets 57% of paths to zero, choosing among many count-equivalent solutions; λ barely matters (between 0.001 and 0.2 it changes held-out count error by 0.32% (March) and 0.25% (April) in an unconstrained-ridge diagnostic and by 0.20% with exact NNLS on D12_I405_S over λ 0.01–0.2); and the hidden reference must be built the same way: a maximum-entropy (KL) projection with an equally good count fit lost 0.03024 (-0.151 S_{ODME}), and bounds from the score formula imply that S_{od} fell by at least 0.288, so the reference is sparse and NNLS-like. **Own-month counts** matter: replacing them by training counts (the official baseline) cost 0.03158 on the public board, which places our S_{ODME} at about 0.994. The counts equal the month's mean flow between 15:00 and 19:00 (ratio 1.000, s.d. 0.013); observed ramp flows cannot serve as extra counts because adding them collapses total path flow by 27–33%.

C. Leaderboard decomposition

Kaggle reports only the total. Attributing every submission's delta to the task it changed (and assuming additivity), V84 is roughly $0.20 S_{\mathrm{ODME}} \approx 0.199$, $0.35 S_{\mathrm{state}} + 0.15 S_{\mathrm{physics}} \approx 0.442$ and $0.30 S_{\mathrm{queue}} \approx 0.262$ ($S_{\mathrm{queue}} \approx 0.874$); the guards of V86 and V87 change only the Task 2 part, by +0.00023 and +0.00032 on the public board. Of the +0.0137 gained from V48 to V84, about two thirds came from the two Task 2 rules (V52, V54) and one third from Task 1 stages. Task 4 hardly separates strong teams, since any team running the official builder on own-month counts obtains the same 0.994.

VIII. Error Analysis and Remaining Headroom

Table VI converts, per task, the remaining loss we can measure into total-score units (IoU differences $\times 0.15$ for one Task 2 condition). Every entry is an offline estimate; none is an official score.

Task 1: blackouts and queue boundaries. On the ruler the official blackouts cost 0.0023 (March) and 0.0027 (April) of total score through S_{state} against a perfect fill, plus about 0.0016 through the LWR proxy, which is optimistic for gap cells (V63). Three rounds of specialists recovered about 0.0008. Ongoing windows hold 59–62% of the blackout cells but 73–75% of the blackout speed error (RMSE about 8 km/h against 5.4–6.2 in onset windows). The states before and after a gap are visible; with demand moving in 30-minute steps, a 90-minute gap can hide up to three demand changes, and the time at which a queue dissolved or spilled back inside it is not identified

by any observation. Outside blackouts, removing the per-link biases and the corridor-wide common mode did not exhaust the free-flow structure: day-level per-link speed offsets (V73g, +0.00031 public) and local context (V76, +0.00030) still paid, and free-flow gains shrank to about 10^{-5} per step only after V78; we have no direct estimate of the noise floor, and the remaining error concentrates on cells where a queue starts or ends between two visible readings. Congested cells were the one class where a density correction transferred fully, once; later refits realised about 40%.

Task 2: block extent, block choice and newborn queues. Onset loss has two parts (pseudo windows, before V54). Picking the wrong block costs 0.066 (March) and 0.037 (April) of panel-mean IoU (production 0.843/0.842 against 0.909/0.879 for an oracle that always picks the best single block), concentrated on D7_I210_W, D12_I5_S and D7_I405_S. The larger part, 0.091/0.121, remains even with the oracle block, because a block's canonical links differ from the links that actually queue. The worst corridor is single-block D7_I10_W, the least congested Task 2 corridor in training (0.36% queued cells), where production equals the oracle at 0.58/0.56: its onsets are marginal, neighbouring links hover at the $0.6\,v_f$ threshold, and the noisy visible speed calls them queued where the latent truth does not (adding such cells cost −0.00250 on the public board, Section IX). For ongoing windows, 283 March replica windows give 0.915 IoU for constrained windows, 0.878 for spacing-bound ones and 0.754 for newborn queues (train-only chain before the selection conditioning), whose IoU falls from 0.91 at T+5 to 0.68 at T+30. Newborn queues are small (median 15 queued cells in the horizon against 44–52 for mature ones) and grow at a rate set by the inflow relative to the bottleneck's discharge, so every misplaced link costs IoU; weighted by the official March class mix (23/8/9 windows), they are 22% of the ongoing windows but carry about 43% of the ongoing loss. March fine-tuning recovers part of it (newborn 0.754 to 0.825).

Task 3 and Task 4. Task 3 costs about 0.03 of total score against an exact answer, mostly in S_{LWR}, which we attribute mainly to unpredictable measurement noise in ΔN (Section VII); Task 4 is the public builder's own-month solve and has at most about 0.0012 of total score left.

What to expect on the private month. Apart from its last sentence, this paragraph gives the expectation we wrote before the private scores were released. The private score depends on components the public board never scored: the April ongoing models fine-tuned on March, the April onset rows, the April instances of every Task 1 stage (whose April-side proxies were positive in all held-out checks) and the April cells of the queue guards (32 in V86, 34 in V87). The selection-rule conditioning realised far more than predicted on March; on April we expect about its prediction (+0.0015 to +0.003, Section VI-D). Tasks 3 and 4 should be stable between the months: real-frame S_{FD} after V61cg (0.9881 vs 0.9880), the Task 4 link fit (0.999939 vs 0.999923) and the zero-path fraction (0.574 vs 0.575) barely differ. By the budget above, the private ranking among strong teams will be decided mainly by Task 2 (newborn ongoing queues and onset block choice; part of the measured onset-extent loss is label noise) and, to a lesser degree, by the Task 1 blackout fill. V86 scored 0.89962 on the private board against 0.90359 on the public one; Kaggle reports only totals, so the difference cannot be attributed to the components above.

IX. What Did Not Work

Negative results were as informative as positive ones. Table VII lists those with a clear outcome; "public" means a change in public score against the build it was applied to.

Lessons.

1) Under a scenario shift, onset choosers that learned where and when congestion usually starts reversed on the corridors with competing bottlenecks; current-state signals relative to each block's breakdown flow transferred, and adapting ongoing models to the new month helped most on newly formed queues.
2) Evaluation structure transfers: the onset step, the origin row and the official selection event produced the largest Task 2 gains that held on the leaderboard.
3) Task 2 input or model changes validated only on pseudo, replica or training months are unreliable at 40 windows per condition; our three largest late Task 2 failures (V43, V67dg, V80) were of this kind.
4) Task 1 proxies transfer their state part almost fully but their conservation part unevenly (0–109%, median about 40%); the first congested-density refill and the per-time-step speed common mode were the two that transferred fully.
5) Small, bounded residual corrections that each pass held-out checks in both scored months accumulated reliably (all eleven late Task 1 steps improved the public score, the last by an attributed +0.00002).

X. Rule Compliance and Disclosures

A. How the final pair complies

- **The final files.** V87 and V86 equal V84, the end of the model chain, except for 104 and 94 Task 2 queue cells set by the queue guards of Section VI-E; their Task 1 and Task 4 rows (and with them the inputs of the Task 3 score) are byte-identical to those of V84.
- **Task 2 inputs.** Every Task 2 model reads only the window history and the origin row T, which the organisers allowed. Onset models, ongoing members and decision rules are fitted on training months (March rows) or on training months plus March (April rows). No April label and no April leaderboard feedback was used. Until V45, April pseudo windows (cut from April's visible layer, with pseudo labels from visible speed) served as

TABLE VI

Remaining loss of V84, the base of the final files, in total-score units (V86 and V87 share its Task 1 and Task 4 rows and differ in 94 or 104 queue cells). "Ruler", "hide", "pseudo" and "replica" are our offline frames; none is an official score. Task 2 IoU differences are multiplied by 0.15 (one condition).

Component	Measurement	Total score	Status
T1 state overall	$0.35(1 - S_{\text{state}})$, S_{state} 0.94–0.95	0.018–0.021	includes the rows below; mostly target measurement noise (assumed); later gains $\approx 10^{-5}$ per step after V78; no direct noise-floor estimate
T1 blackouts	ruler, loss vs perfect fill	0.0031 / 0.0033	0.0039 / 0.0042 before the three specialist rounds (0.0023 / 0.0027 via S_{state}), ≈0.0008 recovered; residual is dissolution timing inside gaps
T1 congested	hide frame, perfect-cell bound	≤ 0.0084	unreachable bound (full LWR proxy); first L1-density fix realised fully, later ≈40%
T2 onset, block choice	pseudo, oracle best block	0.010 / 0.006	measured before V54; D7_I210_W, D12_I5_S, D7_I405_S
T2 onset, block extent	pseudo, oracle block vs truth	0.014 / 0.018	upper bound, includes pseudo-label noise (noisy visible speed, visible cells only); D7_I10_W worst
T2 ongoing	replica, by window class	≈0.02 to perfect	newborn queues 22% of windows, 43% of loss (46% after the selection conditioning, which lowers the loss to 0.018); April models adapted to the scenario
T3 physics	implied by decomposition vs exact answer	≈0.03	S_{FD} saturated (0.988); S_{LWR} gap attributed to measurement noise (hypothesis)
T4 ODME	leaderboard ablations	≤ 0.0012	published NNLS solve; counts fix only cross-section flows

a confirmation gate, also for components that remain in the April rows of V84 and of the final files (disclosure 2). After that, two of our own decisions looked at April visible data that postdate the origin of at least one April window (disclosures 3 and 5); the first kept the train-fitted default and the second only built the hedge file, which was not selected, so neither changed a row of the final files. The hand-off of the V86 guard does not say on which windows its settings were chosen (disclosure 6). Of the queue guards, the V85 guard is causal by its code: a window's query uses only the rows of its day up to and including T, the analogues and their futures come from the training period, before March, and its cached member probabilities come from inputs ending at or before T (by their generating scripts, which are shipped but not re-run). The team member describes the V86 guard only as spatio-temporal gating by neighbouring historical observations. Its code was not available to us when this package was finalised, so we cannot show that it uses only data up to T; we report its 94 cells as an unverified part of the final files (disclosure 6).

- **Truncation checks.** Re-running with every row after each origin deleted changed none of the 87,000 queue cells of the selection-rule conditioning (80 windows) and no onset choice on the four multi-block corridors (a control that also deletes the row at T does flip one window, so the check is not vacuous). An end-to-end truncation check of the queue rows of V84 (`t2_final_asof.py`) re-derives all 160 official windows, each from inputs in which every row after its origin has been deleted (later rows of the day, all later days, later window histories, and every cache derived from them, regenerated inside the truncated view); it covers the April ongoing models, the selection-rule conditioning and all eight onset rules. None of the 174,000 queue cells changed; control runs that also delete the row at T (or every row from $T-5$ on) moved 319 cells, so the check is not vacuous. These checks cover the model-built cells of V84; V86 and V87 differ from them in 94 and 104 guard cells, which no truncation re-run covers: for the ten V85 cells causality rests on the code (above; stage `v85` re-derives them on the full inputs), and for the 94 V86 cells it is not shown (above). The package also rebuilds V84, all of whose Task 2 rows are model-built and pass the truncation check, so the file without the guard cells is available for comparison.
- **Probes.** Sixteen public-only single-window probe files were built (p1–p4 on V48; q1–q3, r1–r5 and s1–s4 on V64); q1–q3 and r1–r5 were submitted, s1–s4 were not, and no submission is recorded for p1–p4. The earlier public-only probe V38 and the diagnostic V11 are listed in Table VIII. No probe edit is part of V86 or V87. The queue rows of V84 equal the model-built V54 March rows and V72 April rows; V86 and V87 differ from them only in the guard cells of their change lists, all in constrained ongoing windows (Section VI-E).
- **Task 1.** Task 1 is offline; using all visible March and April observations, including after the target time, is allowed and is how the month adaptation and calibration stages work.

B. Disclosures

1) **March (public) rows were partly selected with March data after T.** The March blend weights (V47/V48) and several components (the expected-IoU

TABLE VII
IDEAS THAT FAILED, WITH THE EVIDENCE THAT DECIDED THEM.

Task	Idea	What it did	Outcome
T2 onset	month-aware prior (V12)	blended the onset prior with the month's own earlier onsets	public −0.00209 vs V9
T2 onset	expand D7_I10_W onset set (V38)	added two cells that sit at the speed threshold	public −0.00250
T2 onset	block switch on D7_I405_S (V43)	two train-fitted models agreed on the upstream block	public −0.00477
T2 onset	literal pre-registered rule (V54)	would have picked the per-link habit model on three corridors	March replica −0.13 to −0.39 IoU
T2 onset	onset BiLSTM; shockwave features; context/denoising choosers	sequence model or wave features to choose the block	no corridor passed the training gate; IoU 0 on D12_I5_S
T2 ongoing	ml2 alone for ongoing (V15)	replaced the first LightGBM	public −0.00233 vs V8
T2 ongoing	month-to-date queue prior	prior from the month's earlier queues	pseudo 0.834 → 0.79 (blackouts hide queued periods)
T2 ongoing	graph diffusion; group-wise IoU; stricter groups (V29, V34, V35)	alternative smoothing and decision rules	public −0.00001, −0.00005, −0.00012
T2 ongoing	link-ID embedding, kNN analogues, TCN, 3-hour CNN, extra BiLSTM seeds	other members	all weaker than the BiLSTM in pseudo IoU
T2 ongoing	causal within-March fine-tuning for March rows	fine-tune on March days before each window	−0.0036 to +0.0023 by cutoff (unstable)
T2 ongoing	degraded-observation members (V67dg)	fed members the noisier non-eligible observations	public −0.00165
T2 ongoing	trajectory CNN with attention (V80)	new model trained from scratch, positive on training months	public −0.00328
T1	low-rank gap branch in blackouts	low-rank completion inside gaps	8/8 corridors worse on held-out June targets in the 90 min after the training windows' origins (proxy 0.9326 → 0.9248)
T1/T3	L1 density on free-flow cells; refill in blackouts (V63)	extend the congested density fix	dTotal $-1.5 \cdot 10^{-4}$; public −0.00002
T1/T3	LWR smoothing of congested runs; flow common mode; per-(link, day) offsets	further residual structure	$\leq 10^{-6}$, no signal
T1	incumbent prediction as an input feature (V84 candidate)	teacher output fed to the residual model	failed both parity months
T4	maximum-entropy (KL) ODME (V39)	dense solution with the same count fit	public −0.03024
T4	training-month counts (V10)	official baseline	public −0.03158
T4	unconstrained ridge; weighted ridge; ramp counts	alternative regularisers and extra counts	≈28% negative flows; held-out 2–6% worse; flow collapse

decision of V33, the CatBoost shares, the V30 random forest for D7_I405_N) were chosen on pseudo windows cut from the whole March month; the V52 decision to exclude spacing-bound windows and the V54 exclusion of habit families were taken after seeing March replica results. A public-only probe file proposing the D12_I5_S window change later made by V54 had been built on 19 October; no submission of it is recorded. These choices affect public rows only, not the private ranking.

2) **April pseudo windows were used as a gate until V45 (18 October), three days after the 15 October rulings, which we applied from V46 on.** From the earliest onset work until V45, April pseudo windows served as a confirmation gate (V41–V45 on 17–18 October still report April pseudo gains), and April pseudo numbers were still reported (not used to choose) for V46–V48. Components from that period remain in the April chain: the train-only co-queue groups, the parameter-free expected-IoU decision and the onset choosers of V18/V24. After that, April-specific choices (V50–V72) were made on March data, except the two decisions of disclosures 3 and 5, which also looked at April visible data.
3) **April onset on D12_I5_S.** The decision to keep the train-fitted onset rule for April rather than a March-derived habit was informed by April visible data before the window origins (queues on block 43–45 before the origins of windows 002–005, which postdate the origin of window 001). It kept the default rule, so no April row differs from a train-only build.
4) **V71 deviated from its pre-registered rule**: the nested half-month check failed in one direction and the team adopted the change anyway; V72, whose April rows V84 carries, passed its own rule.
5) **The registered hedge concerns a file that was not selected:** its rule, written after seeing the chooser probabilities and pre-origin visible data of April D12_I5_S onset windows that postdate the origin of the one window it switches, built `v84_hedge.zip`, which is not a final file; neither V86 nor V87 contains its switch (Section VI-E).
6) **Final queue guards.** The V85 gate (at least 20 valid analogues, queue-share margin 0.30) was chosen among several rules on 251 March proxy windows that had been used repeatedly, and its intervals contain zero (Section VI-E); for its eight March cells this is selection with March data after T, as in disclosure 1; for its

two April cells it is a use of March, which the rules allow. The code of the V86 guard is not in the package: its 94 decisions (62 March and 32 April cells in 18 windows; 63 set from queued to not queued, 31 from not queued to queued) ship as a frozen change list, which the build applies to the rebuilt V84 after checking each cell's recorded previous value, and we know its method and inputs only from the team member's description. Because the list is applied as recorded, a re-run with the rows after each origin deleted reproduces these 94 cells by construction and says nothing about their causality. The hand-off for the final pair notes that local queue evaluation used proxy labels from visible observations and the same windows repeatedly to select rules, so those evaluations are not an independent test. V87 was assembled after V85 and V86 had been submitted (its file is 20 minutes younger than theirs), so their public scores may have informed the combination.

7) **Post-hoc candidates.** For V73–V79 some candidates (clipping variants, a flow bound, a boosting step size) were added after first results on the same folds; from V81 on, amplitudes were chosen on parity and frozen.
8) **Frozen artifacts.** Four model files used only by the parity gates of V81/V82 were produced by an earlier revision of the code and are shipped as frozen inputs; they do not affect the final values. Historical Task 2 decisions of V48/V51 are shipped as reference files against which the rebuild asserts. Unlike these, the 24 CPU probability caches that the V85 guard reads decide its ten cells; they ship frozen in the team member's hand-off (their generating scripts are included but not run), and the V86 decisions ship as a frozen list (disclosure 6).

C. External data, models and licenses

No external data was used: only the official release (training layer, the masked validation and private months, corridor networks and configuration), whose license statement reads "Subject to the Competition Rules"; its network topology is derived in part from OpenStreetMap, whose data is available under the ODbL. No pretrained model from outside the project and no API or paid tool were used; every network and booster was trained by us on the release. The Task 4 reference builder is the organisers' public code (MIT License). All libraries are open source: numpy, pandas, scipy, scikit-learn and PyTorch (BSD), LightGBM (MIT), pyarrow, CatBoost and XGBoost (Apache-2.0); the report figures use matplotlib (PSF-based license). The CUDA 12.1 build of PyTorch comes with NVIDIA's CUDA and cuDNN runtime libraries, which are freely redistributable but not open source. Our code is released under the MIT License, and the package README lists every random seed behind the submitted values.

XI. Reproducibility

Package. The code package (https://github.com/vacucity/IEEE-Big-Data, about 1.1 GiB) keeps the project layout and adds `README.md`, `LICENSE`, `MANIFEST.json` (SHA-256 of every file), the report PDF, `work/requirements.txt`, `work/ENVIRONMENT.txt`, copies of the submitted V87 and V86 in `work/reference/final/`, the team member's V85 hand-off in `work/v85_history_guard/` and one entry script, `work/run_all.py`. Its stage table (`--plan`) gives, for each stage from data cleaning to V86 and V87, the commands, outputs, GPU need, verification status and the runtime measured in our clean copy; `--list-frozen` lists the shipped artifacts each stage reads; `--run-all` and `--from` run the stages in order, logging every command, and after an interruption `--resume` continues stage by stage, skipping the stages that already ended without error (an interrupted stage restarts from its first command once its partial outputs are deleted; see the README); after the clean-copy run below all 36 stages are marked `verified` (`--force`, needed for stages marked `todo` or `partial`, is no longer required). `--check` compares the SHA-256 of every output with the result manifests, which a fresh copy writes during its own run (a consistency check, not a comparison with our build), and, with `--ref-root`, compares outputs by content with a reference tree. The last stage, `verify`, compares V87 and V86 with the SHA-256 of the submitted files and line by line with the copies in `work/reference/final/` (or in a `--ref-root` tree), and passes if either matches. In the development tree, whose V84 is the original file, both files built by stage `final` are byte-identical to the submitted ones; in a rebuilt tree the zips from V60 on hold the same content as ours but different line ends on 521 lines (the submitted V84, V86 and V87 end 725 lines in LF and the rest in CRLF, their rebuilds 204; two state-only intermediate zips differ on 570), so there V86 and V87 match by content (see Verification below). The official release (11.5 GiB unzipped) goes into `kaggle_public/`; everything not shipped is regenerated, which needs about 45 GB of free disk besides the release (our clean copy held 38 GB of regenerated files at the end). The package ships the trained networks and boosters that inference needs (56 files in `work/models`, about 1.0 GiB, plus the three V30 stack boosters and the four V81 parity-gate boosters under `work/results`), the frozen decision artifacts (selected configurations, registered rules, gate decisions, historical Task 2 decisions and queue files that the rebuild is checked against, the two guard change lists), the V85 hand-off (its code, 24 frozen CPU probability caches and 8 training-period co-queue tables) and the official Task 4 reference files; the smaller LightGBM and ridge models of the calibration stages are refitted inside their stages. Optional retraining commands (`--retrain`) exist only for the blackout gap U-Nets of stages `t1_`

v59blk (4 gap-specialist and 8 self-supervised nets) and t1_v61b3 (8 nets) and for the four April members of t2_v72 fine-tuned on March with 96k windows; every other shipped model, among them the production Task 1 U-Nets and base boosters, the other Task 2 members and the two April BiLSTMs fine-tuned with 48k windows, has no training command in the stage table.

Task 2 from models. t2_final_queue.py writes the five Task 2 queue files (two for March, three for April) in one pass from the member models, using the per-corridor onset rules and ongoing recipes of the original patch chain; the files are byte-identical to the originals (rerun with the final code on 2 November). t2_v71_april.py --variant v72 then writes the April ongoing rows that V84 carries. The check modes of t2_final_onset.py and t2_final_ongoing.py re-derive all 160 official windows from the models with zero mismatching cells. t2_final_asof.py repeats this with every row after each window's origin deleted (0 of 174,000 cells change; Section X); these checks form stage t2_check, which runs after V84 is built because t2_final_asof.py first matches the final queue files with the queue rows of V84.

Final files. Stage v85 (build_v85_history_guard.py) writes the 174,000 queue rows of the rebuilt V84 and asserts that they equal the rows the team member used, runs the guard's own audit script from work/v85_history_guard/ on clean_data/ and kaggle_public/, asserts that its three audit files equal the originals (line ends ignored), and re-derives the 10 cells, asserting that they equal the official V85 change list and the V85 rows of the V87 list; the shipped script passed in 85 s in the development tree (original V84) and in 88 s in the clean copy (rebuilt V84). The 24 CPU probability caches are frozen inputs (their historical generating scripts are in the hand-off's reference/ folder and are not run); the hand-off's copies of our modules are identical to ours, except that evaluate.py reads the data paths from the environment. The team member tested the guard with numpy 2.3.5 and pandas 2.3.3; stage v85 passed with the versions below. Stage final (build_final_v86_v87.py) writes V86 as V84 plus the frozen V86 change list and V87 as V84 plus that list and the recomputed V85 cells, asserted equal to the frozen V87 list; for every flipped cell it asserts that the value in V84 equals the recorded previous value, copies every other byte of V84 unchanged, and writes each zip as the submitted ones were written (one member submission.csv, deflate, 1980 timestamp). --list-frozen lists the V86 change list as a frozen input, because the code of that guard is not part of the package. Stage verify then compares both files with the submitted ones (above).

Environment. Python 3.11.5 with numpy 1.26.4, pandas 2.0.3, pyarrow 22.0.0, scipy 1.10.1, scikit-learn 1.7.2, LightGBM 4.6.0, CatBoost 1.2.8, XGBoost 2.1.3 and PyTorch 2.5.1 (CUDA 12.1). The U-Net, CNN and BiLSTM inference stages need a CUDA GPU with bf16 support; all other stages run on the CPU. The reference machine is a laptop with an Intel Core i7-13700H (14 cores, 20 threads), 16 GB RAM and an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB, driver 555.97) under Windows 11 (build 26200). On Windows run_all.py keeps the system and the display awake (under Modern Standby a screen-off timeout can pause processes; it paused one clean-copy run for 6.3 h) and ignores Ctrl+C; a closed lid, Sleep or sign-out still interrupts the run. The pinned scikit-learn version matters because one onset rule is a seeded random forest. Table XI lists the main model settings and Table X the lineage of the final files with SHA-256 prefixes. The stages of a full run take 18.2 h in total on the reference machine without retraining (stage times measured in the clean copy, interruptions excluded), 3.6 h of it in the six GPU stages; the Task 1 calibration stages (9.3 h, the per-month bias stage t1_v52 alone 4.4 h) and the late Task 1 stages (5.0 h) dominate, the three Task 2 stages take 40 min and v85, final and verify 6 min.

Determinism and tolerance. Every random draw is seeded in code (the README lists the seeds behind the submitted values). The LightGBM models refitted inside the stages run in deterministic mode (deterministic=True); on the default path the shipped networks and boosters are only used for prediction, and inference from the shipped networks is deterministic on a given GPU. Retraining a shipped model is not guaranteed to be bit-exact: the BiLSTM uses non-deterministic cuDNN kernels, and the Task 1 base boosters and the Task 2 LightGBM members were trained with fixed seeds but without deterministic mode. On a different GPU, bf16 rounding could flip individual cells of the top-k and Monte-Carlo decisions. A flip in a final cell would be absorbed by the 1% tolerance, but most Task 2 cells are checked exactly: the April rebuild asserts that the recomputed V48 and V51 decisions equal the recorded ones (stage t2_queue checks V51, t2_v72 both), and t2_check compares the five queue files cell by cell with the shipped originals, so by default a flip there stops the run with an error. The guard cells are checked exactly as well: v85 asserts its recomputed audit and cells, and final the previous value of every flipped cell. For a GPU model other than ours, run_all.py --tolerant (or TFB_TOLERANT=1) turns exactly these comparisons with recorded values into reported differences: each prints the number of differing cells, the Task 2 rows keep the values recomputed on that GPU, the guard stages apply the recorded guard decisions, and verify reports the differing lines; structural checks and the as-of truncation check stay fatal. We tested the mode by altering recorded decisions and two V84 cells on purpose: every check reported the difference and the run continued; the Task 2 files stayed byte-identical, the guard cells took their recorded values, and the final files differed from the submitted ones only in the altered V84 cell outside the guard lists. Eight builders originally asserted the exact byte hash of their input zip; because

a rebuilt zip holds the same values in different bytes even on the development machine (the mixed line ends above; another platform can add further differences), these checks now warn instead of stopping unless `TFB_STRICT=1`, while every value-replay assertion inside the builders still applies. Reverting this change restores source files whose hashes equal those recorded in the build manifests, which shows that nothing else was modified. The award rule requires the reproduced private score within 1% (about 0.009); Task 2 rules are nevertheless rebuilt cell-exactly.

Verification. A clean copy of the package next to the official release, without any output of the development tree, ran all 36 stages from 4 to 8 November (`--run-all --resume --force --ref-root` with the development tree as reference) and ended with `verify` passing: the rebuilt V87 and V86 are content-identical to the submitted files (6,980,504 lines each); their SHA-256 differs because 521 lines end in CRLF where the submitted files have LF, the same lines as in the rebuilt V84. Along the way every stage from `t1_base` on compared its outputs with the development tree; the 40 cleaned parquet files were compared by SHA-256 after the run, and stage `t4` compared the rebuilt Task 4 reference with the shipped copy. These, every state and queue file and every zip still kept in the development tree (V51, V60–V84) are byte- or content-identical, as are the refitted LightGBM models and the ridge coefficients of V81/V82; the twelve intermediate zips of V52–V59 are no longer kept there and were checked through their state files and the later zips. `t2_check` again found 0 mismatching cells in the 160 windows and 0 of 174,000 cells changed by the truncation check, and `v85` recomputed the guard's audit and its 10 cells on the rebuilt V84. Among the compared outputs, the only byte differences are timings and absolute paths recorded in result files, the SHA-256 of the rebuilt V84 in the `v85` manifest, one optional diagnostic column that only the development run wrote, and floating-point noise of at most $1.2 \cdot 10^{-13}$ in the coefficients of V73/V74. The run was resumed after interrupted consoles and after five stops on gaps in the package that we then closed: a missing working folder, `t2_check` scheduled before V84 existed, a missing feature-alignment audit for V77 (whose first version stopped the run once more because it counted NaN in the same cells as unequal), and stage `final` reached before stage `v85`, added during the run, had written its output; a 6.3 h standby pause only delayed it. No stage script was changed after its stage had passed. An earlier clean room (25–26 October) had reproduced the cleaned data, the production Task 1 chain and the first calibration stages.

XII. Conclusion

TrafficFlowBench rewards understanding the generator and the evaluation more than model capacity. Our final files, V86 and V87, are the last build of our model chain, V84, with 94 and 104 of its 174,000 Task 2 queue cells flipped by queue guards (Section VI-E); all their other rows are those of V84. V86, the better of the two on the private leaderboard, scored 0.89962 (7th of 802 teams on the Kaggle private leaderboard). Three lessons carry beyond this benchmark. First, under a scenario shift the transferable signal is the current state relative to physical limits (flow against each bottleneck's breakdown flow, the queue's present extent), while onset choosers that learned where and when congestion usually starts reversed on the corridors with competing bottlenecks; validation that cannot see the shift (leave-one-month-out on training months) must be complemented by out-of-scenario rulers. Second, evaluation mechanics are part of the problem: blackouts that blank whole corridors, onset at a fixed step and published window-selection rules each produced either the largest loss or the largest transferable gain, and modelling them explicitly beat generic improvements. Third, the value of an offline proxy lies in knowing its bias: state proxies transferred almost fully, conservation proxies unevenly, and Task 2 proxies only when they encoded the official window structure.

With more time we would attack the remaining budget in this order: newborn ongoing queues, which carry the largest share of the ≈0.02 ongoing loss; onset block extent (0.014/0.018 for March/April on pseudo windows, an upper bound that includes pseudo-label noise; worst on the single-block D7_I10_W) and block choice on D7_I210_W, D12_I5_S and D7_I405_S (0.010/0.006), both measured before V54; the dissolution timing of queues inside blackout gaps (≈0.003), which no visible cell identifies and which therefore needs a generative model of queue lifetimes; and joint modelling of density for Tasks 1 and 3 at congested cells.

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Appendix A
Public-Leaderboard History

Tables VIII and IX list every scored file with a logged score; they are generated from the submission log by `final_package/report/make_tables.py`.

Appendix B
Lineage of the Final Files

Each late build patches the previous file; Table X gives the chain with file hashes. Queue and Task 4 rows of every build from V72 to V84 are identical to V72; V80 branches off V79 and is not in the lineage. The two selected files differ from V84 only in Task 2 queue cells set by the queue guards (Section VI-E): V86 in 94 cells, V87 in the same 94 plus the 10 cells of V85 (disjoint, 104 in total); their Task 1 and Task 4 rows are byte-identical to V84. V85 (V84 with 10 queue cells, b594280f) was submitted on its own and not selected; its 10 cells are part of V87; a registered hedge file (V84 with 4 April queue cells, 3bbb0beb) was built but not selected.

Appendix C
Main Hyper-parameters

Appendix D
Glossary

TABLE VIII

EVERY SUBMISSION WITH A LOGGED SCORE, IN LOG ORDER (PUBLIC LEADERBOARD = MARCH 2031). "VS BEST": DIFFERENCE TO THE BEST SCORE BEFORE IT. ^{f} ONE OF THE TWO SELECTED FINAL FILES; ^{r} BELOW THE BEST AT THE TIME (REJECTED); ^{p} PUBLIC-ONLY PROBE OR DIAGNOSTIC, NEVER PART OF A FINAL SUBMISSION; ^{d} ATTRIBUTION TEST OR BASELINE COMPARISON; ^{c} COMPONENT TEST, ADOPTED IN THE NEXT BUILD (V31 WAS SCORED ON THE V28 QUEUE ROWS, +0.00012 AGAINST V28, AND CONFIRMED AS V32). BUILDS WITHOUT A LOGGED NUMERIC SCORE ARE OMITTED: UNSUBMITTED BUILDS, APRIL-ONLY DUPLICATES AND THE PUBLIC-ONLY PROBES Q1–Q3 (EACH REPORTED BELOW 0.900). PROBE SCORES R2–R4 COME FROM THE TEAM NOTES. THE V84 SCORE IS ATTRIBUTED FROM THE TEAM BEST; THE V85–V87 SCORES ARE FROM SCREENSHOTS OF THE KAGGLE SUBMISSION PAGE (7 NOVEMBER, UTC+8). V85–V87 CHANGE ONLY TASK 2 QUEUE CELLS OF V84; V86 DOES NOT CONTAIN THE V85 CELLS, SO ITS "VS BEST" ENTRY COMPARES TWO DIFFERENT CHANGE SETS. AGAINST V84 THEY GAIN +0.00009, +0.00023 AND +0.00032 (NOT ADDITIVE, SECTION VI-E). OTHER TRIALS BY THE TEAM MEMBER WHO BUILT THEM ARE NOT IN THE LOG. (PART 1 OF 2)

Build	Date	Public	vs best	Task	Change
V2	2026-10-06	0.70223		all	first complete file: baseline state, rule-based queue, own-month ODME
V3	2026-10-06	0.82699	+0.12476	T1	temporal/spatial interpolation for state
V4	2026-10-07	0.83449	+0.00750	T1	LightGBM residual state model
V7^{r}	2026-10-09	0.83430	-0.00019	T2/T4	ongoing gap fill; NNLS ODME restored
V8	2026-10-09	0.84431	+0.00982	T2	per-link LightGBM queue models (onset and ongoing)
V9^{d}	2026-10-09	0.84395	-0.00036	T2	LightGBM for ongoing only, onset rules (attribution test vs V4)
V10^{d}	2026-10-09	0.81273	-0.03158	T4	NNLS on training counts (official baseline)
V11^{p}	2026-10-10	0.74378	-0.10053	T2	diagnostic: empty onset predictions
V12^{r}	2026-10-10	0.84186	-0.00245	T2	month-aware onset prior
V13	2026-10-11	0.84559	+0.00128	T2	onset threshold 0.55
V14	2026-10-11	0.84994	+0.00435	T2	regularised ml2 LightGBM for onset
V15^{r}	2026-10-11	0.84198	-0.00796	T2	ml2 for ongoing
V18	2026-10-11	0.86854	+0.01860	T2	onset as block choice: flow logit
V19	2026-10-12	0.87395	+0.00541	T1	whole-day U-Net blended with LightGBM
V20	2026-10-12	0.87543	+0.00148	T1	three-way blend incl. conservation-loss U-Net
V21	2026-10-12	0.87723	+0.00180	T2	ongoing ensemble LightGBM + ml2 + CNN
V23	2026-10-12	0.87904	+0.00181	T1/T2	width-48 U-Nets; CNN 3 seeds
V24	2026-10-13	0.88026	+0.00122	T1/T2	month adaptation; capacity-distance onset
V25	2026-10-13	0.88044	+0.00018	T2	corridor BiLSTM member
V26	2026-10-13	0.88188	+0.00144	T2	BiLSTM 2 seeds, reweighted
V27	2026-10-14	0.88205	+0.00017	T2	ongoing weights rebalanced
V28	2026-10-14	0.88219	+0.00014	T2	train-only co-queue group smoothing
V29^{r}	2026-10-14	0.88218	-0.00001	T2	graph diffusion smoothing
V30	2026-10-14	0.88469	+0.00250	T2	D7_I405_N set-level onset forest
V31^{c}	2026-10-14	0.88231	-0.00238	T1	residual stack on 4 corridors
V32	2026-10-15	0.88481	+0.00012	T1/T2	V30 queue + V31 state
V33	2026-10-15	0.88538	+0.00057	T2	expected-IoU top-k decision
V34^{r}	2026-10-15	0.88533	-0.00005	T2	group-wise IoU decision
V35^{r}	2026-10-15	0.88526	-0.00012	T2	stricter co-queue groups
V36	2026-10-15	0.88547	+0.00009	T1	stack transferred to 5 more corridors
V37	2026-10-16	0.88547	+0.00000	T1	half-strength stack on D7_I210_E
V38^{p}	2026-10-16	0.88297	-0.00250	T2	D7_I10_W onset set expansion
V39^{r}	2026-10-16	0.85523	-0.03024	T4	maximum-entropy (KL) ODME
V40	2026-10-16	0.88549	+0.00002	T1	half-strength stack on 2 corridors
V41	2026-10-17	0.88572	+0.00023	T2	CatBoost 10% on 2 corridors
V42	2026-10-18	0.88577	+0.00005	T2	CatBoost 20% on D7_I10_E
V43^{r}	2026-10-18	0.88100	-0.00477	T2	D7_I405_S onset block switch
V44	2026-10-18	0.88589	+0.00012	T2	CatBoost 30% on D7_I10_E
V45	2026-10-18	0.88914	+0.00325	T2	CatBoost on D7_I405_S (score log ambiguous)
V46	2026-10-18	0.88914	+0.00000	T2	origin row T used (shifted inputs)
V47	2026-10-18	0.88936	+0.00022	T2	members retrained T-anchored (origin row)

TABLE IX
EVERY SUBMISSION WITH A LOGGED SCORE (PART 2 OF 2, CONTINUED FROM TABLE VIII).

Build	Date	Public	vs best	Task	Change
V48	2026-10-19	0.88971	+0.00035	T2	+ T-anchored BiLSTM seed (final March blend)
V50	2026-10-19	0.88971	+0.00000	T2	April only: March fine-tuned members
V51	2026-10-19	0.88971	+0.00000	T2	April only: more fine-tuned seeds
V52	2026-10-19	0.89520	+0.00549	T2	conditioning on the selection event
V53	2026-10-19	0.89606	+0.00086	T1	per-month per-link bias
V54	2026-10-20	0.89981	+0.00375	T2	replica-ruled onset rules
V56	2026-10-20	0.90005	+0.00024	T1	per-link band refinement
V57	2026-10-20	0.90022	+0.00017	T1	month-specific restack
V58	2026-10-20	0.90059	+0.00037	T1	+ blackout residual model
V60	2026-10-21	0.90101	+0.00042	T1	local features; gap-specialist U-Net
V61	2026-10-22	0.90159	+0.00058	T1	congested L1-density refill
V62	2026-10-22	0.90167	+0.00008	T1	+ blackout specialist average (E1)
V63^{r}	2026-10-23	0.90165	-0.00002	T1	density refill in blackouts
V64	2026-10-23	0.90173	+0.00006	T1	congested model pooled over months
R1^{p}	2026-10-23	0.90173	+0.00000	probe	public-only onset probe (D12_I5_N)
R5^{p}	2026-10-23	0.90173	+0.00000	probe	public-only onset probe (D12_I5_S)
R2^{p}	2026-10-23	0.89798	-0.00375	probe	public-only onset probe (D7_I210_W)
R3^{p}	2026-10-23	0.89798	-0.00375	probe	public-only onset probe (D7_I210_W)
R4^{p}	2026-10-23	0.89798	-0.00375	probe	public-only onset probe (D12_I5_S)
V67^{r}	2026-10-24	0.90058	-0.00115	T1/T2	common mode + degraded-observation members
V67dg^{r}	2026-10-24	0.90008	-0.00165	T2	degraded-observation members alone
V66	2026-10-24	0.90223	+0.00050	T1	corridor-wide speed common mode
V69	2026-10-24	0.90228	+0.00005	T1	second stage on common-mode residual
V71	2026-10-25	0.90228	+0.00000	T2	April only: 48k-window fine-tunes
V71h	2026-10-25	0.90228	+0.00000	T2	April only: + hedge window
V73g	2026-10-27	0.90259	+0.00031	T1	visible-data speed factors, clipped
V74	2026-10-27	0.90261	+0.00002	T1	low-rank speed factors
V75	2026-10-27	0.90277	+0.00016	T1	LightGBM factor mixer
V76	2026-10-27	0.90307	+0.00030	T1	context speed mixer + flow residual
V77	2026-10-28	0.90316	+0.00009	T1	77-feature flow model
V78	2026-10-28	0.90320	+0.00004	T1	flow model +350 trees
V79	2026-10-28	0.90321	+0.00001	T1	flow model +300 finer trees
V80^{r}	2026-10-28	0.89993	-0.00328	T2	trajectory CNN with attention
V81	2026-10-28	0.90329	+0.00008	T1	congested speed common mode
V82	2026-10-28	0.90333	+0.00004	T1	congested interaction ridge
V83	2026-10-30	0.90334	+0.00001	T1	flow model +300 trees
V84	2026-10-31	0.90336	+0.00002	T1	observation-only flow residual (end of the model chain)
V85	2026-11-06	0.90345	+0.00009	T2	history guard: 10 queue cells of V84 (not selected)
V86^{f}	2026-11-06	0.90359	+0.00014	T2	spatio-temporal guard: 94 queue cells of V84
V87^{f}	2026-11-06	0.90368	+0.00009	T2	V86 + the 10 V85 cells (104 queue cells of V84)

TABLE X
LINEAGE OF THE SELECTED FILES V86 AND V87 (SHA-256 PREFIXES OF THE SUBMISSION FILES).

Build	Base	SHA-256	Rows changed
V72	V69	6115b36a	55 April queue cells
V73g	V72	5b9e0ee7	6.24M free speed
V74	V73g	49e0cba8	6.20M free speed
V75	V74	6f1efe90	6.22M free speed
V76	V75	9186d308	6.09M speed, 6.34M flow
V77	V76	b68f1d20	6.33M free flow
V78	V77	32aa511b	6.33M free flow
V79	V78	83df8bf6	6.31M free flow
V81	V79	785c1d37	269k congested speed
V82	V81	a4b4676d	261k congested speed
V83	V82	ddab5075	6.29M free flow
V84	V83	a429599a	6.32M free flow
V86 (selected)	V84	9e1ba052	94 queue cells (guard)
V87 (selected)	V86	6bc4e383	10 queue cells of V85

TABLE XI
MAIN MODEL AND GUARD SETTINGS (FIXED SEEDS; THE SETTINGS AND SEEDS OF THE V86 GUARD ARE NOT AVAILABLE, AS ITS CODE IS NOT IN THE PACKAGE).

Model	Settings
T1 LightGBM	46 features; speed, per-lane flow and L1-density residual regressors; 3833 / 1460 rounds at learning rate 0.05 (speed, flow) and 2198 rounds at 0.1 (density); trained on a 4% sample of training targets
T1 U-Net	4-level residual 2-D U-Net, width 48 (7.29M parameters), 23 input channels, 30 epochs per fold, 60% of days with 1–2 corridor-wide 90-min gaps, 8% random hiding, conservation weight 0.05 (one of two nets); month adaptation 6 epochs at 10^{-4}
T2 BiLSTM	GRU over each link's history, 2-layer bidirectional LSTM along the corridor, 0.75M parameters; BCE + 0.5 soft IoU; AdamW, one-cycle schedule; 12–20 epochs of 12,000 windows
T2 CNN	corridor 1-D U-Net, width 96, three seeds
T2 ml2	LightGBM; ongoing: learning rate 0.045, 31 leaves, depth 8, min. 250 samples per leaf, L2 8, feature/bagging fraction 0.85; onset: learning rate 0.04, 15 leaves, min. 100 per leaf, L2 10, feature fraction 0.9
March fine-tune	half training / half March windows per batch, learning rate $3{\cdot}10^{-4}$ one-cycle, 24k–96k windows, March history hidden at a random rate up to 20%; half-month cross-fitting
T2 conditioning	20,000 Monte-Carlo samples, acceptance floor 0.02, seed CRC32 of the window id, dissolve rule on D7_I210_E from 17:00
T2 guard (V85)	history guard on constrained ongoing windows, not where the dissolve rule applies or acceptance is below 0.02; query: the 12 steps up to T in at most 16 spatial bins (speed, queue share, coverage); 30 nearest training-period origins (before 2031-03-01) within ±60 min; decode $0.9\,q_{\text{cpu}} + 0.1\,q_{\text{hist}}$ and q_{cpu} and apply the difference to the CPU replay (first step fixed); keep only proposed $0 \to 1$ flips where V84 also has 0, $n_{\text{valid}} \geq 20$ and $q_{\text{hist}} - q_{\text{cpu}} \geq 0.30$
T2 guard (V86)	spatio-temporal guard, shipped as a frozen list of 94 cells (4 ambiguous flips blocked by the build); settings not available (code not in the package)
T4	NNLS with ridge to the prior, $\lambda = 0.05$, own-month counts (official builder)

TABLE XII
TERMS USED IN THIS REPORT.

Term	Meaning
panel, corridor	one directional freeway section, e.g. D12_I5_S
position	link index in milepost order
VN, VNx [VM]	build N; a stage built separately (suffix) and first scored inside file M
suffixes	t1: Task 1 part; rs: restack; blk: blackout; loc: local features; cg: congested; g: guarded; dg: degraded observations; h: hedge; E1: specialist average
pseudo window	window cut from a month's visible layer with simple rules; labels from visible speed
replica window	window chosen by our re-implementation of the official selector
hide-20% frame	20% of visible cells hidden again and reconstructed, giving labelled pseudo targets in a scored month
real frame	the actual submission targets (Task 3 S_{FD} can be computed there)
blackout ruler	977 simulated official-like blackouts in March/April used to score gap fills
held-out checks	parity days, half months and opposite month, in both scored months
LWR proxy	Task 3 conservation term with the true accumulation change replacing the withheld boundary flows
block	recurrent bottleneck: adjacent links that switch to congestion together
ml2	window-weighted, regularised per-link LightGBM
T-anchored	a model trained to forecast from a history ending at the origin row T
constrained window	ongoing window not bound by the 360-min spacing whose $T-5$ origin was rejected only for persistence
spacing-bound window	ongoing window whose origin lies exactly 360 min after the previous window's origin, so $T-5$ was excluded by the spacing
newborn window	ongoing window whose queue formed shortly before T
habit	a prior on when or where congestion occurs, learned from training months
queue guard	a rule that flips individual Task 2 queue cells of V84 (V85, V86; V87 unites both change sets); no model is retrained